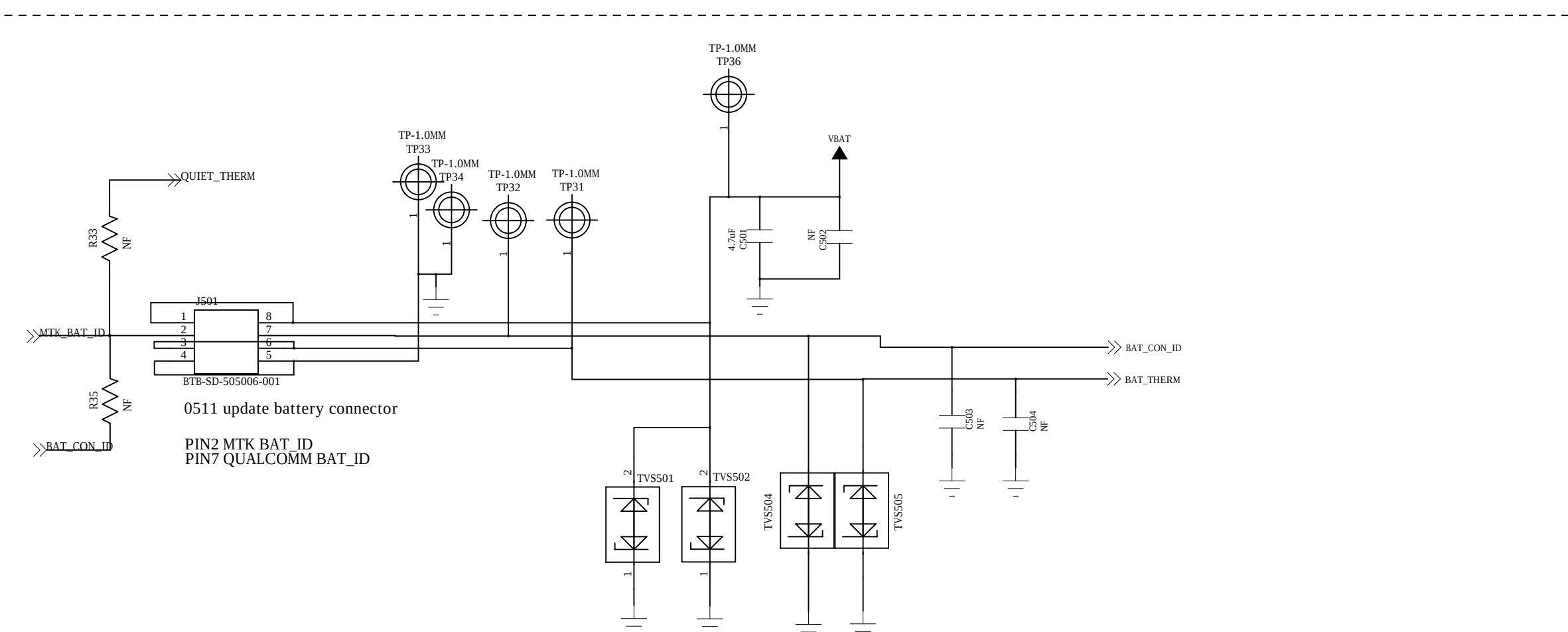
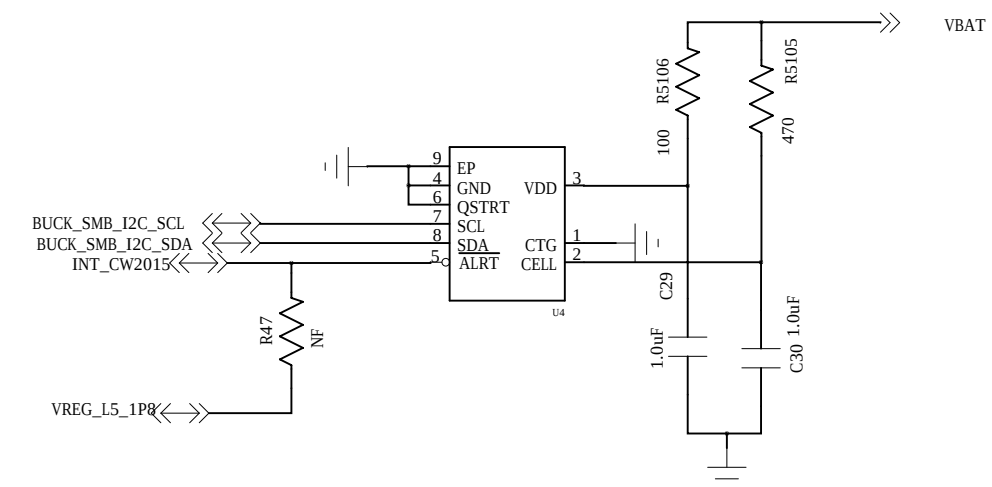
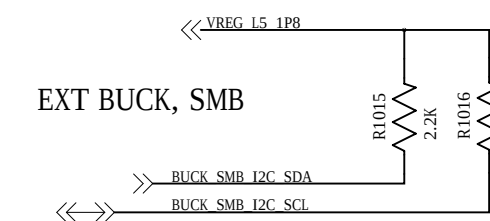
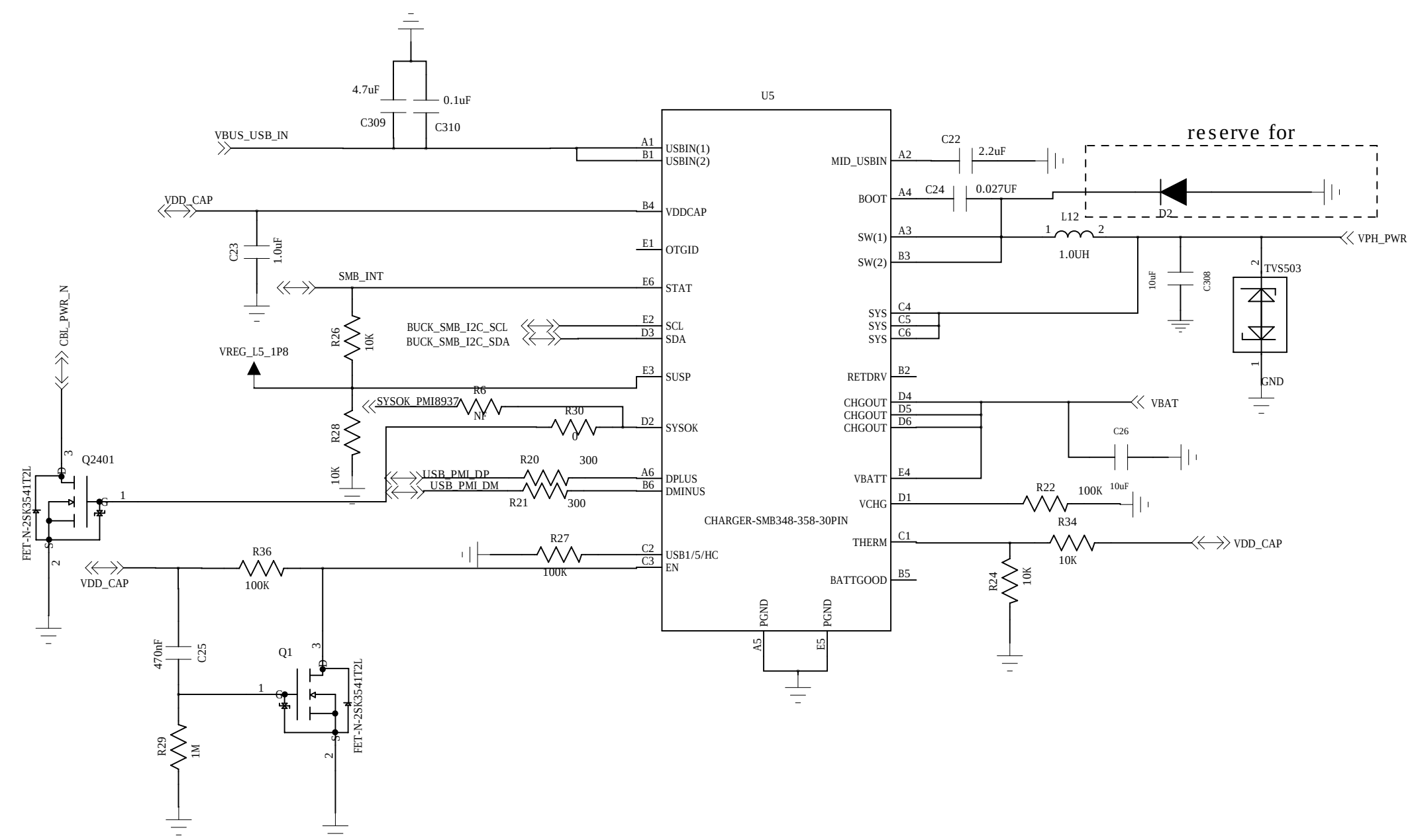




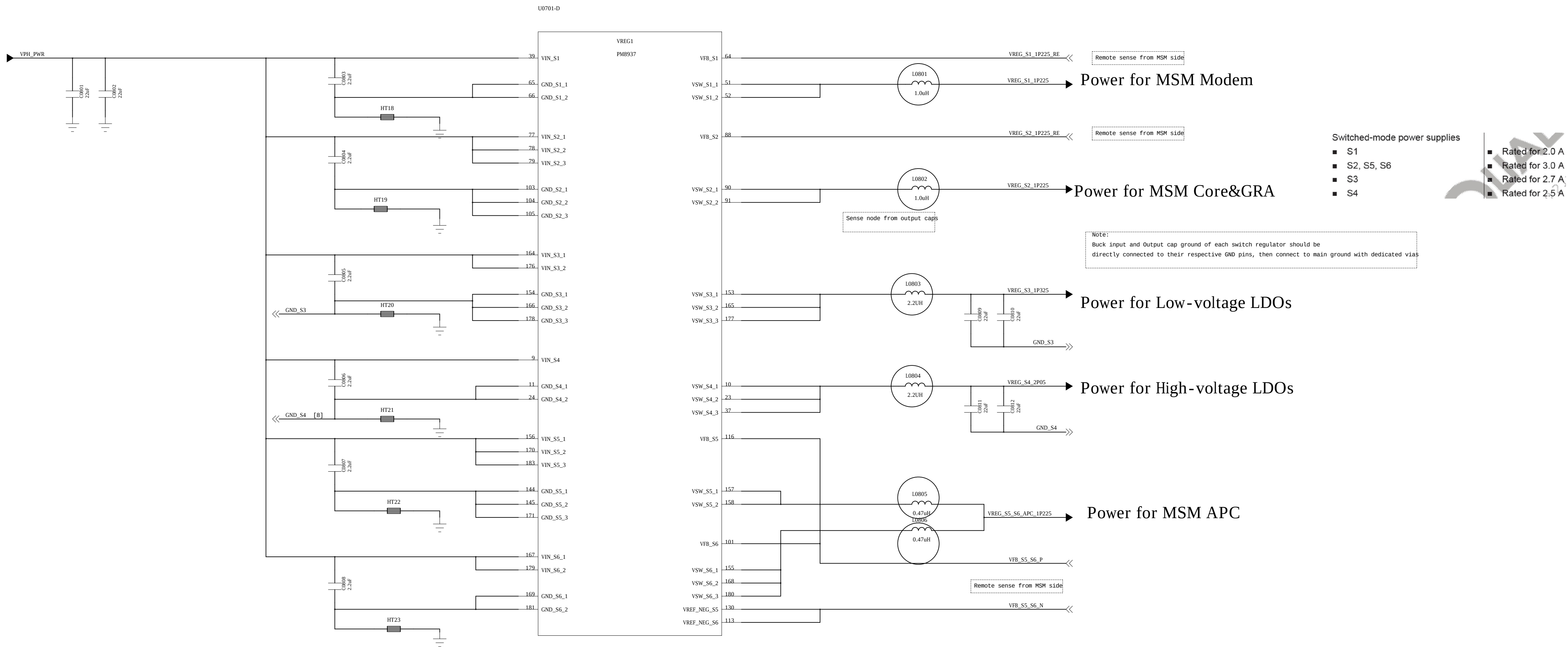
BATID Resistor Pull down to GND in BAT Module.

BAT_THERM NTC(68K B=4250) Pull down to GND in BAT Module.

BATTERY CONNECTOR

PMI8937 CHARGER

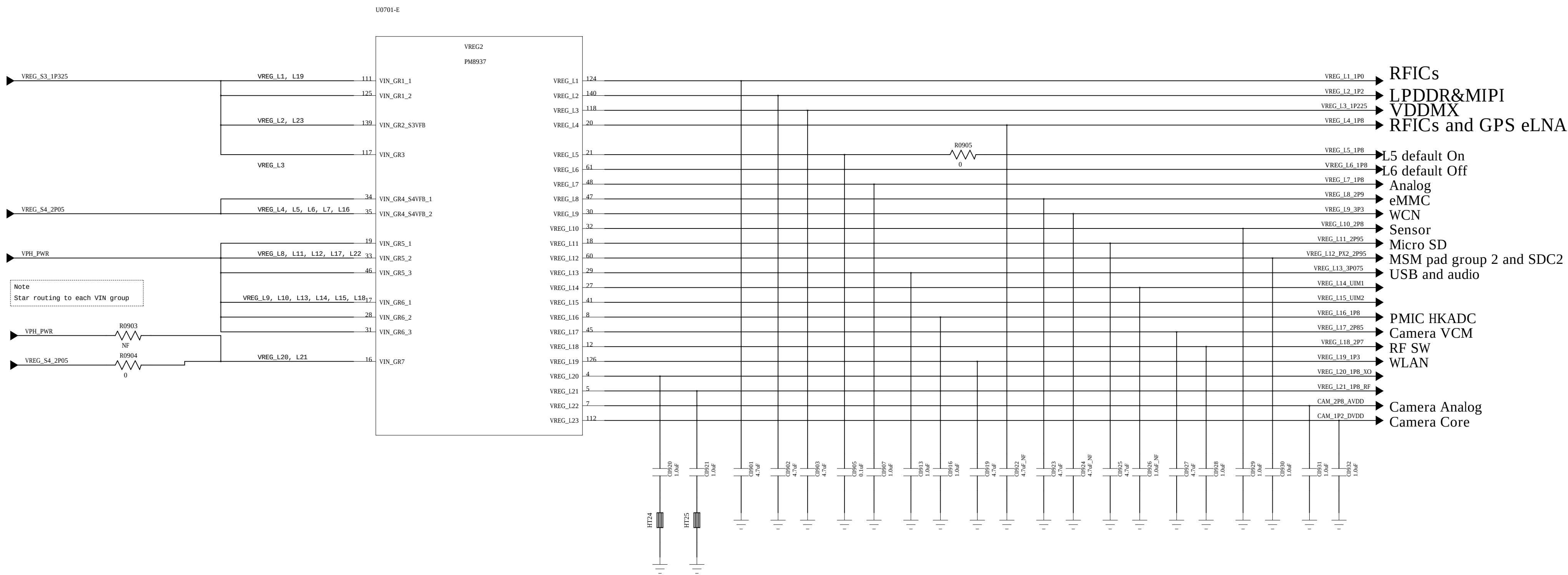
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PM8937 Buck converter

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PSEUDO CAPLESS LDOS

L4*, L6, L8*, L9*, L10, L11*, L12, L14, L15, L17*, L18, L22, L23

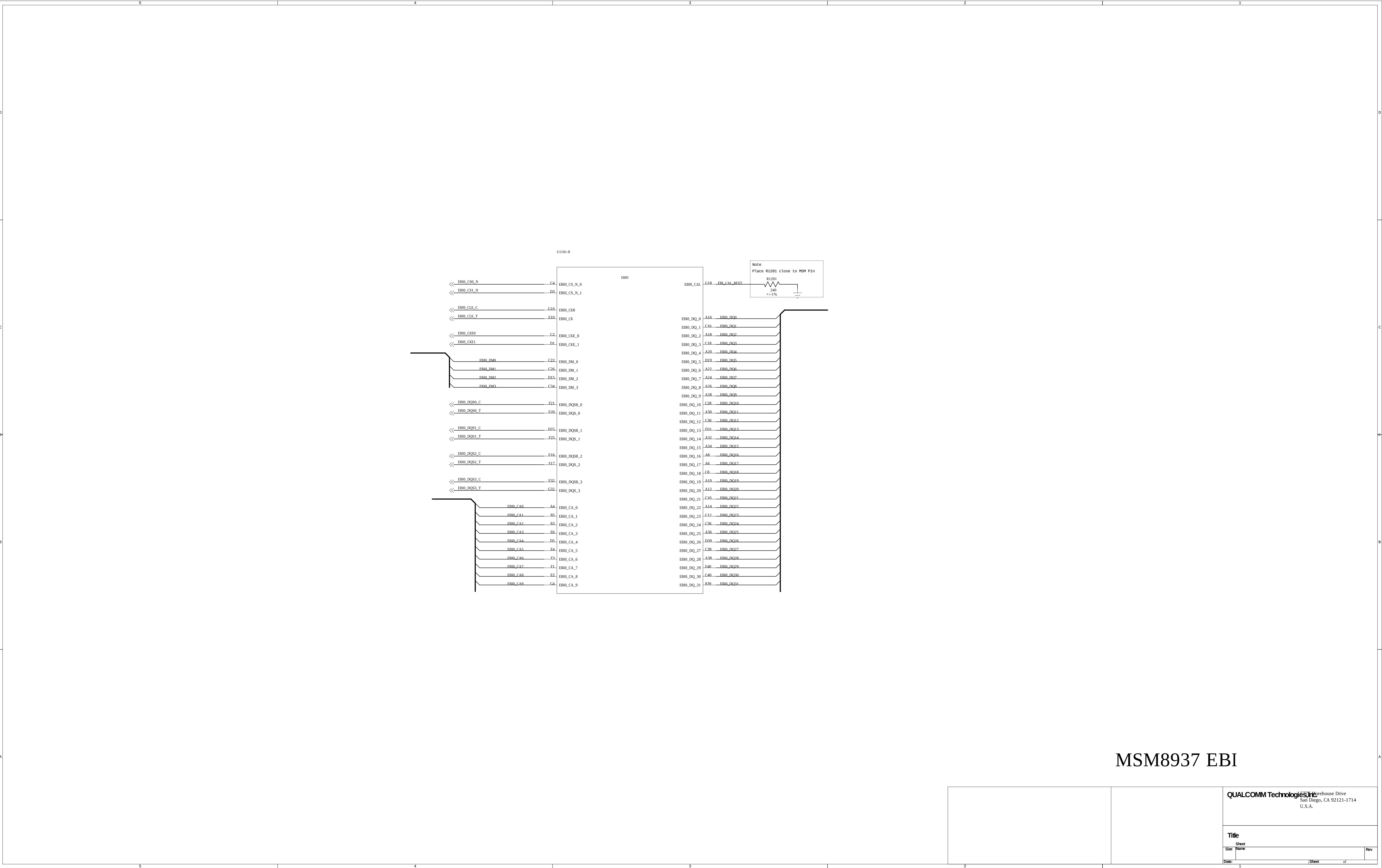
L4*, L8*, L9*, L11*, L17*, still need validation as Pseudo-capless

L5, L7, L13, L16 have internal PMIC Loads and require local CAPS.

PM8937 LDO circuits

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54321

EBI

EBI0_CS0_N

EBI0_CS1_N

EBI0_CLK_C

EBI0_CLK_T

EBI0_CKE0

EBI0_CKE1

EBI0_DM0

EBI0_DM1

EBI0_DM2

EBI0_DM3

EBI0_DQS0_C

EBI0_DQS0_T

EBI0_DQS1_C

EBI0_DQS1_T

EBI0_DQS2_C

EBI0_DQS2_T

EBI0_DQS3_C

EBI0_DQS3_T

EBI0_CA0

EBI0_CA1

EBI0_CA2

EBI0_CA3

EBI0_CA4

EBI0_CA5

EBI0_CA6

EBI0_CA7

EBI0_CA8

EBI0_CA9

EBI0_CS_N_0

EBI0_CS_N_1

EBI0_CK0

EBI0_CK1

EBI0_CKE_0

EBI0_CKE_1

EBI0_DM_0

EBI0_DM_1

EBI0_DM_2

EBI0_DM_3

EBI0_DQS0_0

EBI0_DQS_0

EBI0_DQS0_1

EBI0_DQS_1

EBI0_DQS0_2

EBI0_DQS_2

EBI0_DQS0_3

EBI0_DQS_3

EBI0_CA_0

EBI0_CA_1

EBI0_CA_2

EBI0_CA_3

EBI0_CA_4

EBI0_CA_5

EBI0_CA_6

EBI0_CA_7

EBI0_CA_8

EBI0_CA_9

EBI0_CAL

EBI0_CAL_EXT

EBI0_DQ0

EBI0_DQ1

EBI0_DQ2

EBI0_DQ3

EBI0_DQ4

EBI0_DQ5

EBI0_DQ6

EBI0_DQ7

EBI0_DQ8

EBI0_DQ9

EBI0_DQ10

EBI0_DQ11

EBI0_DQ12

EBI0_DQ13

EBI0_DQ14

EBI0_DQ15

EBI0_DQ16

EBI0_DQ17

EBI0_DQ18

EBI0_DQ19

EBI0_DQ20

EBI0_DQ21

EBI0_DQ22

EBI0_DQ23

EBI0_DQ24

EBI0_DQ25

EBI0_DQ26

EBI0_DQ27

EBI0_DQ28

EBI0_DQ29

EBI0_DQ30

EBI0_DQ31

EBI0

EBI0_CAL

EBI0_CAL_EXT

EBI0_DQ0

EBI0_DQ1

EBI0_DQ2

EBI0_DQ3

EBI0_DQ4

EBI0_DQ5

EBI0_DQ6

EBI0_DQ7

EBI0_DQ8

EBI0_DQ9

EBI0_DQ10

EBI0_DQ11

EBI0_DQ12

EBI0_DQ13

EBI0_DQ14

EBI0_DQ15

EBI0_DQ16

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EBI0_DQ22

EBI0_DQ23

EBI0_DQ24

EBI0_DQ25

EBI0_DQ26

EBI0_DQ27

EBI0_DQ28

EBI0_DQ29

EBI0_DQ30

EBI0_DQ31

Note

Place R1201 close to MSM Pin

R1201

240

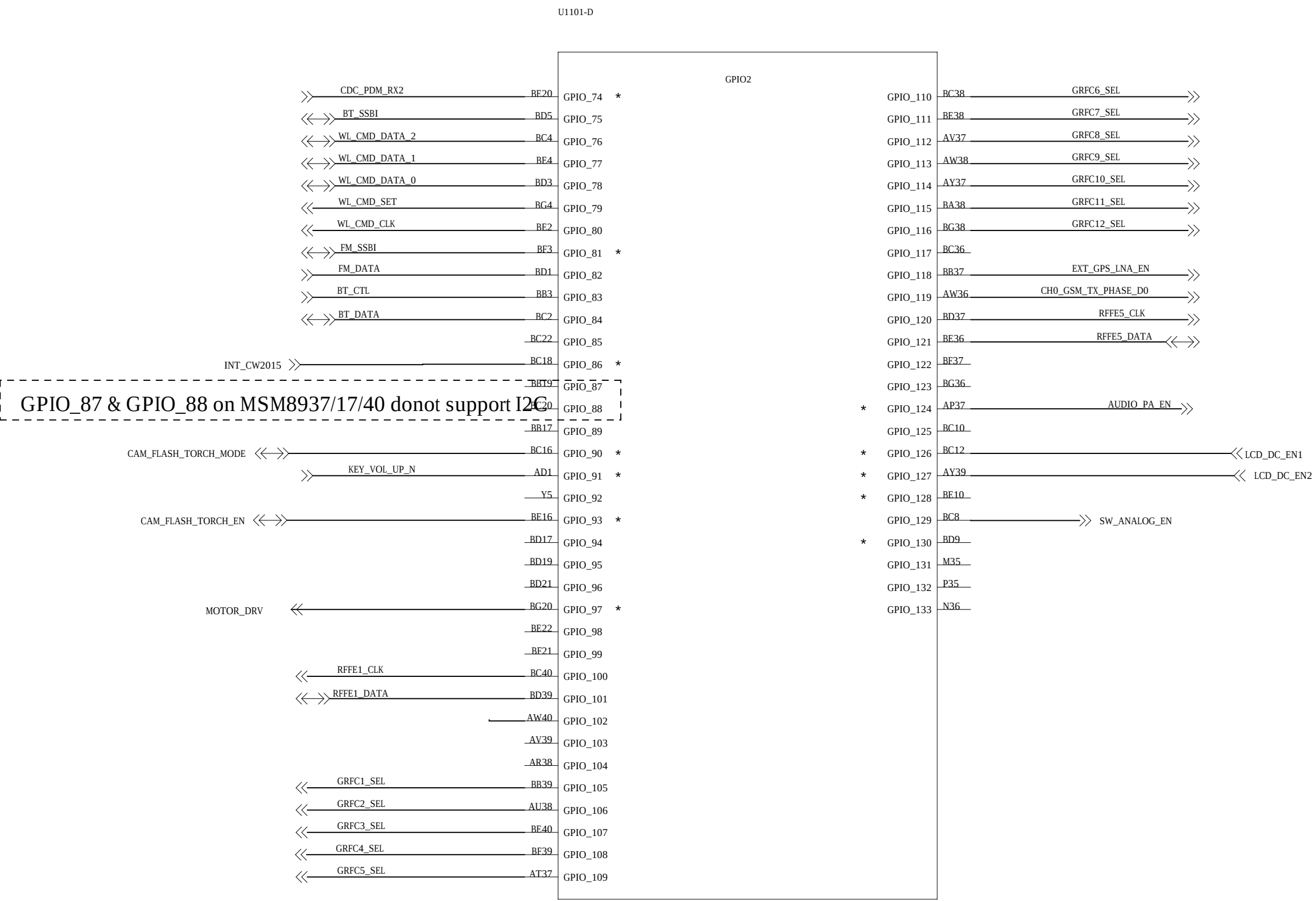
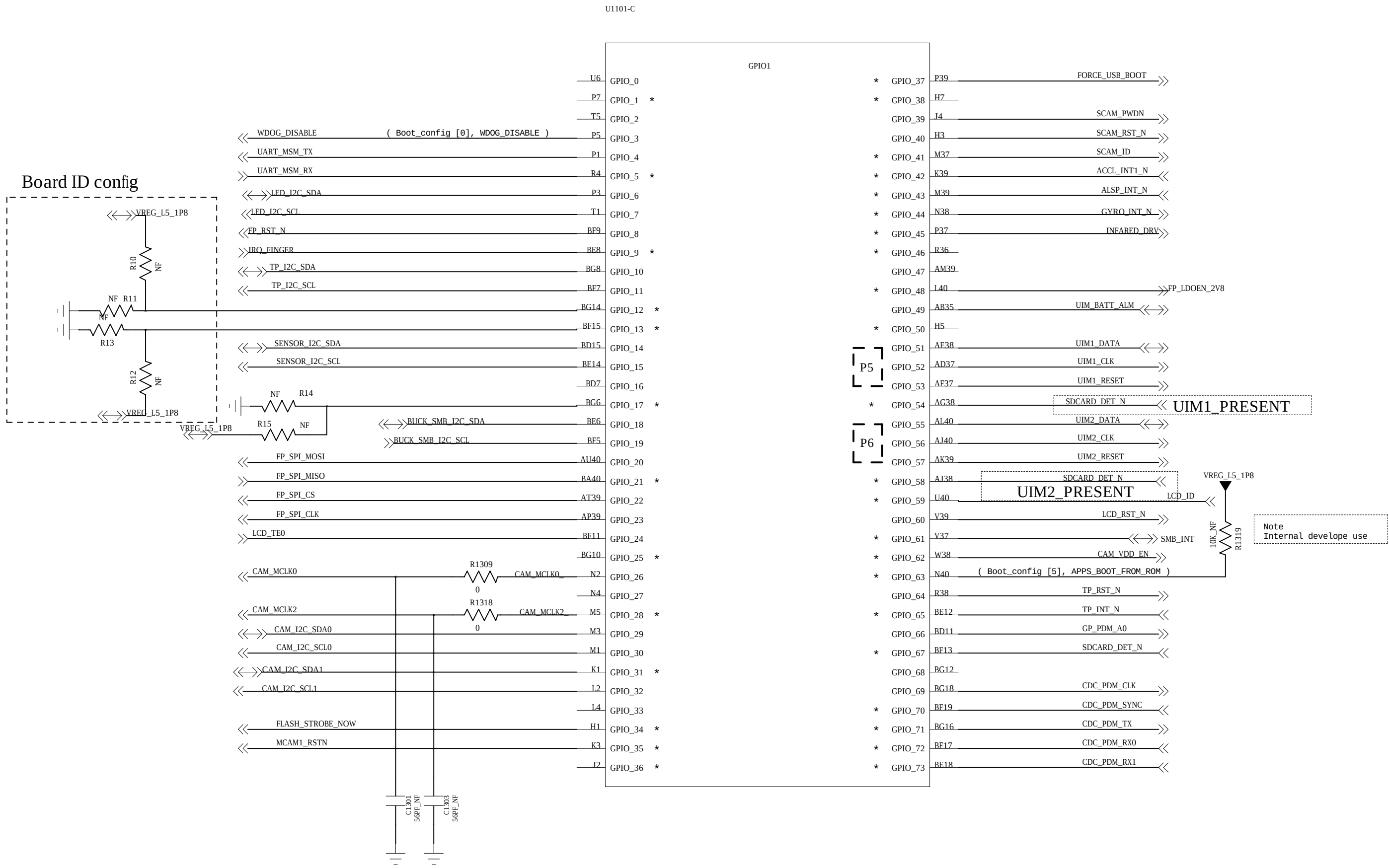
+/-1%

MSM8937 EBI

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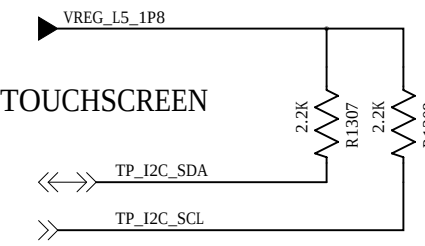
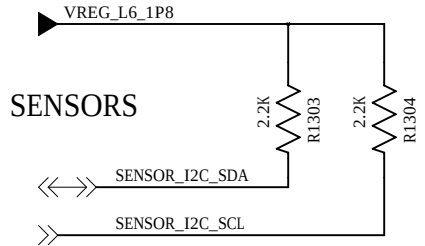
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Sheet		
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Date:	1	Sheet of

Note: Asterisks (*) indicate modem power management (MPM) wake-up pins

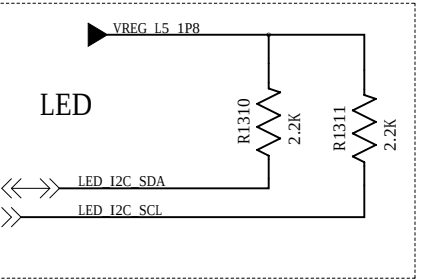
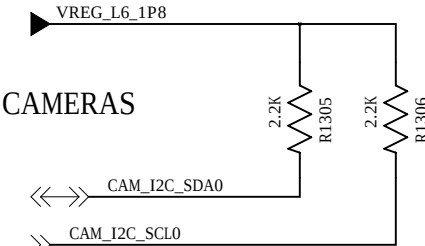


Note: Do not have pull-ups on the following GPIOs unless intend for boot or secure- boot related configurations:
GPIO_91, 107, 109

I2C PULL-UP RESISTORS



Note: Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to inout pull down when the peripherals are powered off to eliminate leakage.



Depth CAMERAS

MSM8937 GPIO

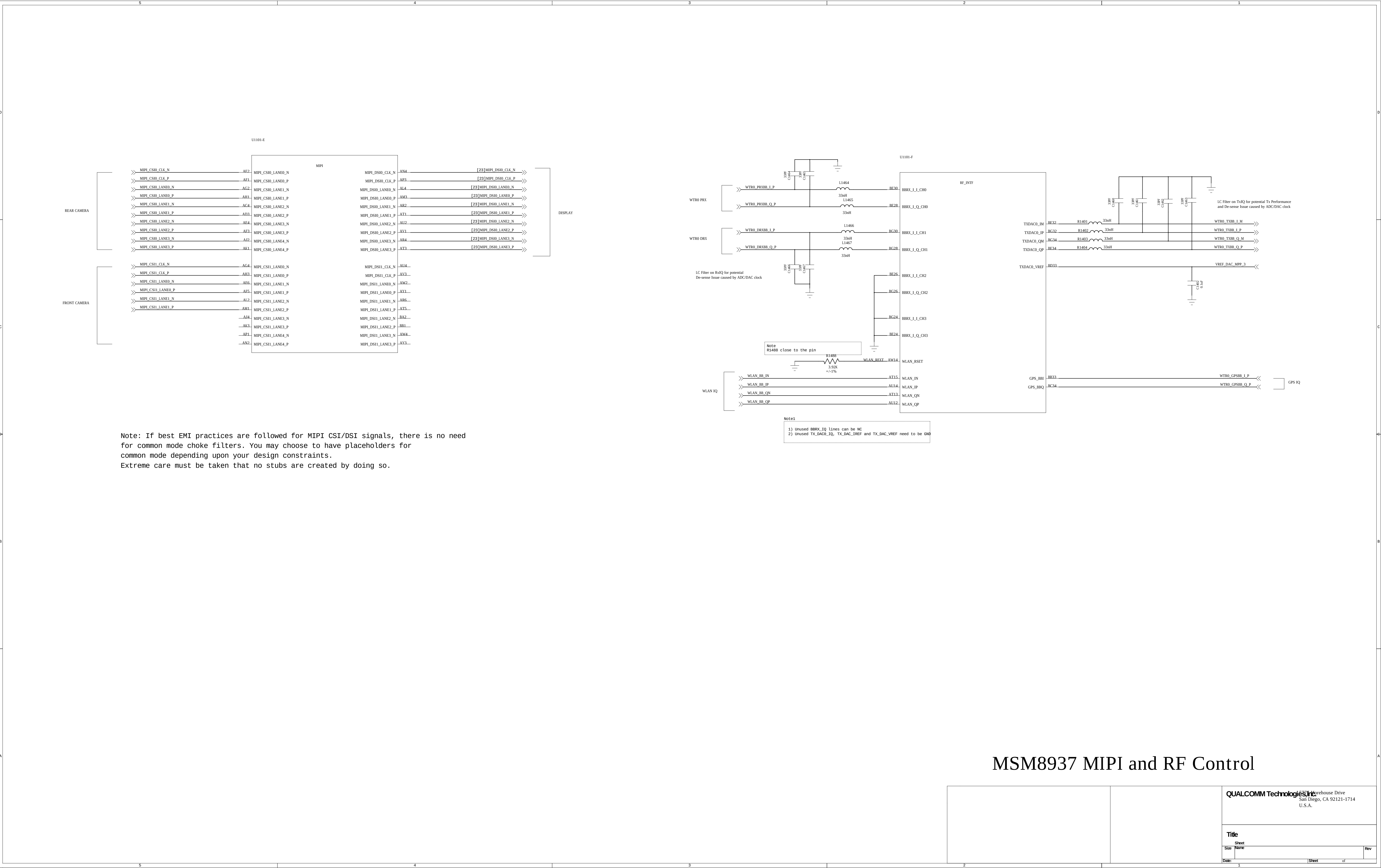
GPIO	BOOT_CONFIG
GPIO_3	BOOT_CONFIG[0]/WDOG_DISABLE
GPIO_111	BOOT_CONFIG[1]
GPIO_112	BOOT_CONFIG[2]
GPIO_88	BOOT_CONFIG[3]

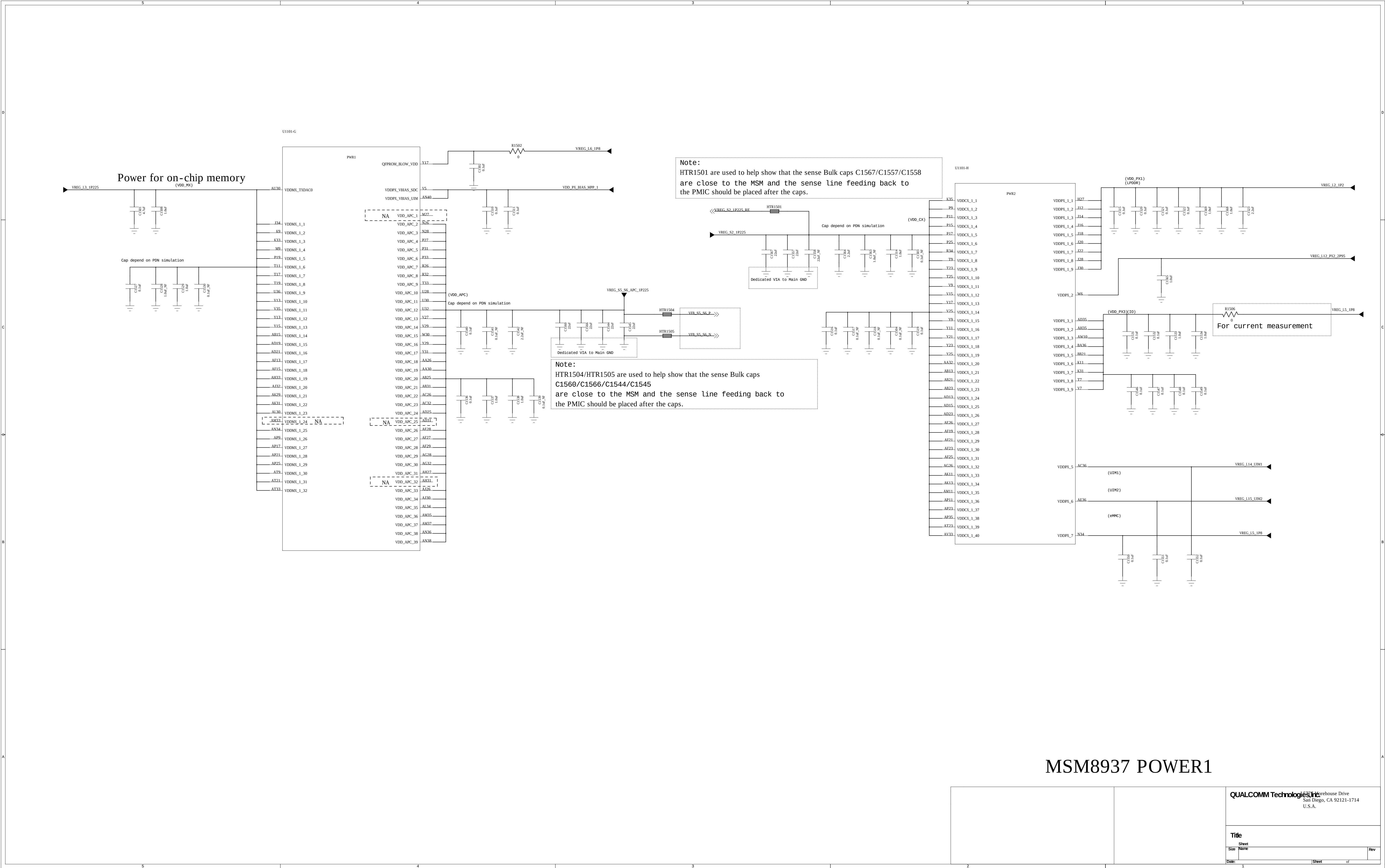
BOOT_CONFIG[3:1]	BOOT_CONFIG
0b000	SOC1 -> SOC2 -> USB2.0
0b010	SOC1
0b100	SOC2-> SOC1

Default Boot Config (0b000) is SOC1(eMMC)

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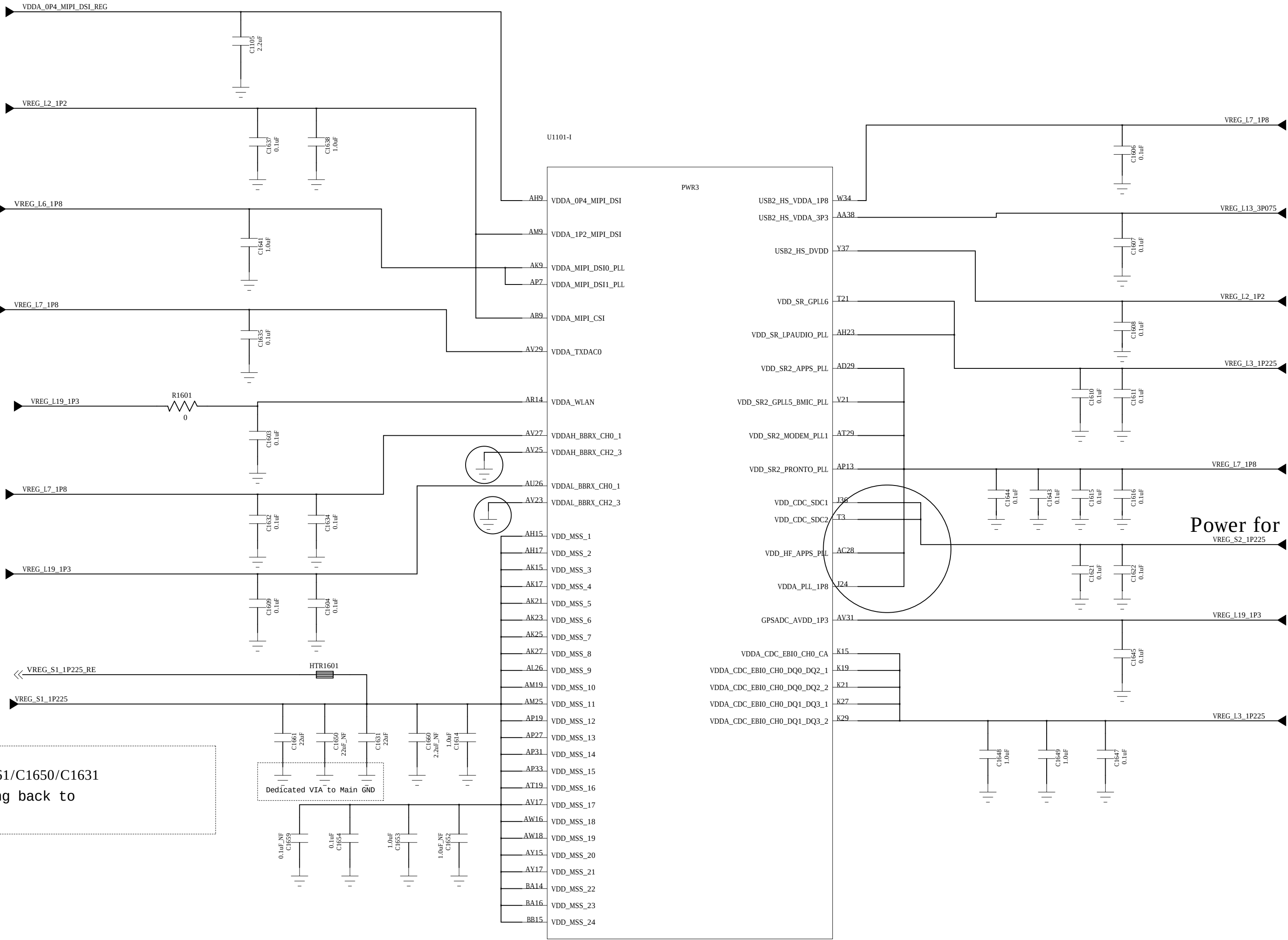


MSM8937 POWER1

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Note:
HTR1601 are used to help show that the sense Bulk caps C1661/C1650/C1631 are close to the MSM and the sense line feeding back to the PMIC should be placed after the caps.

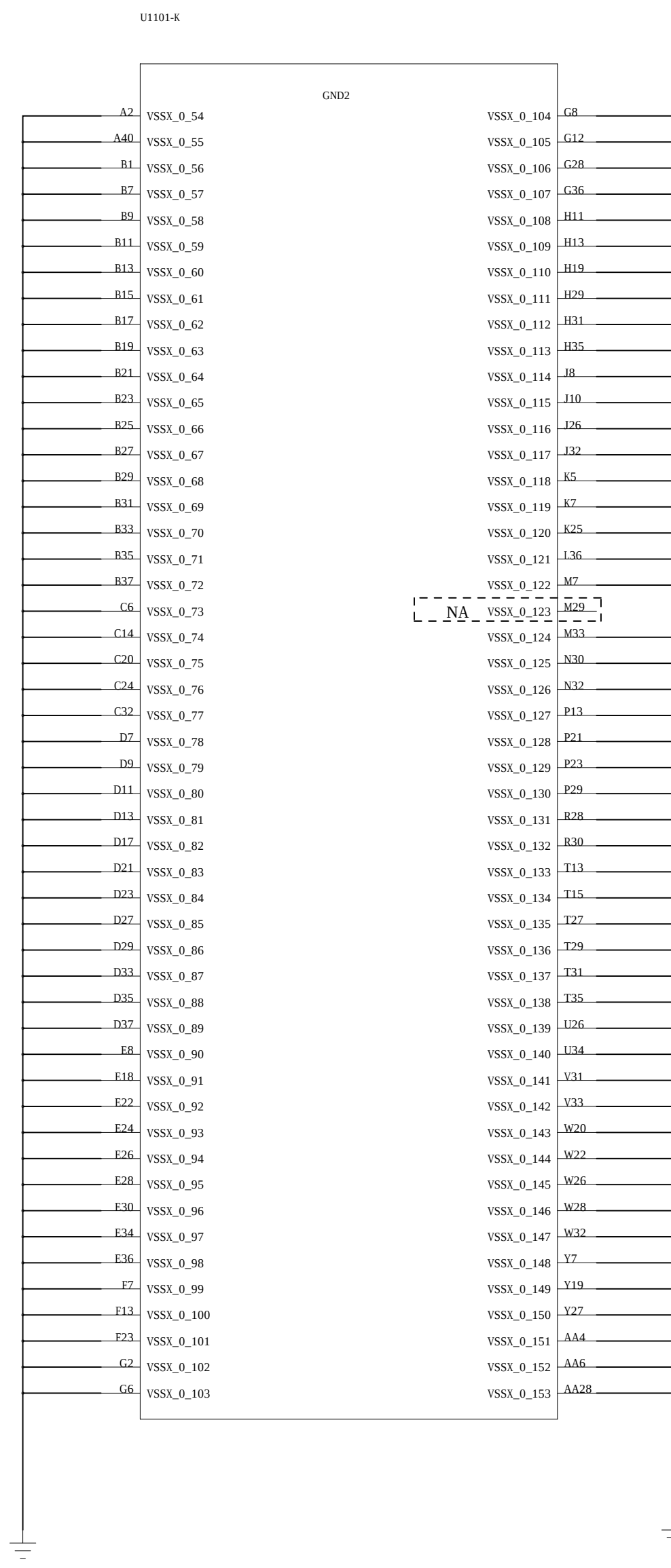
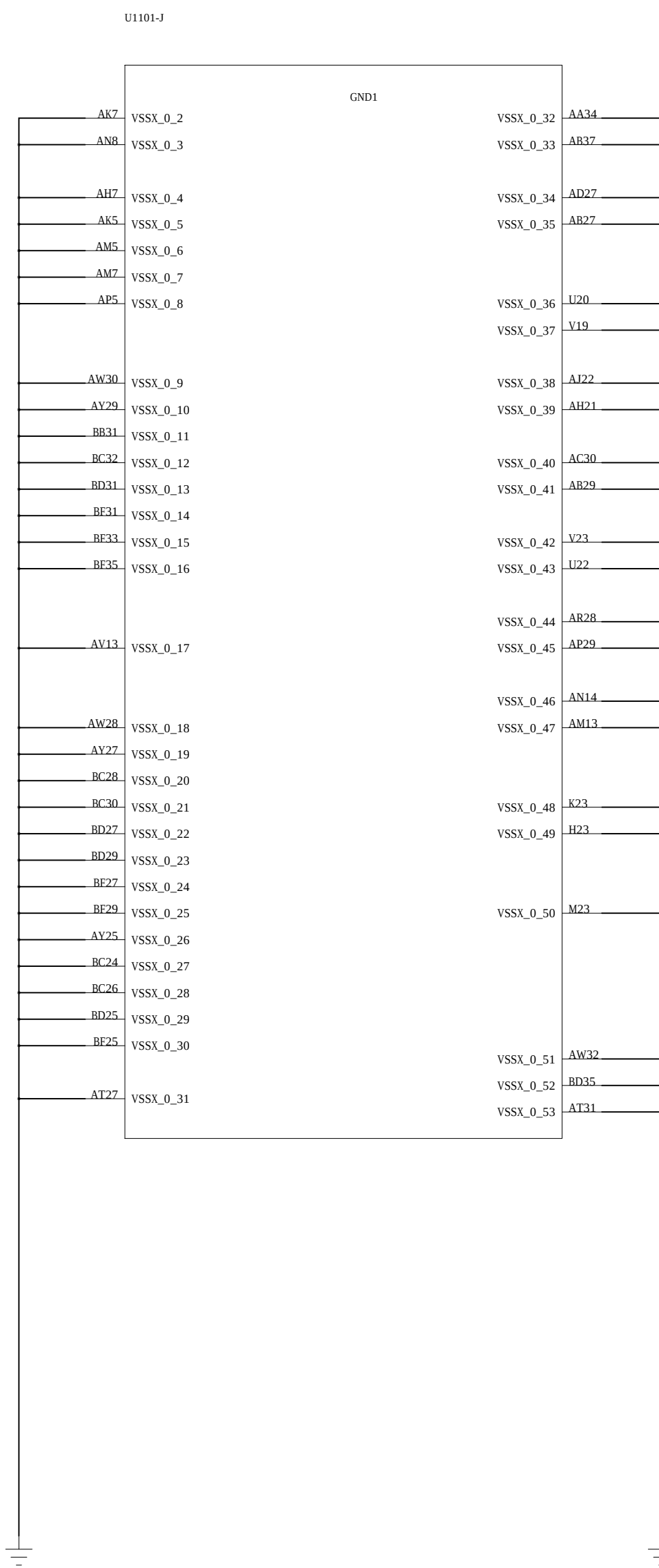
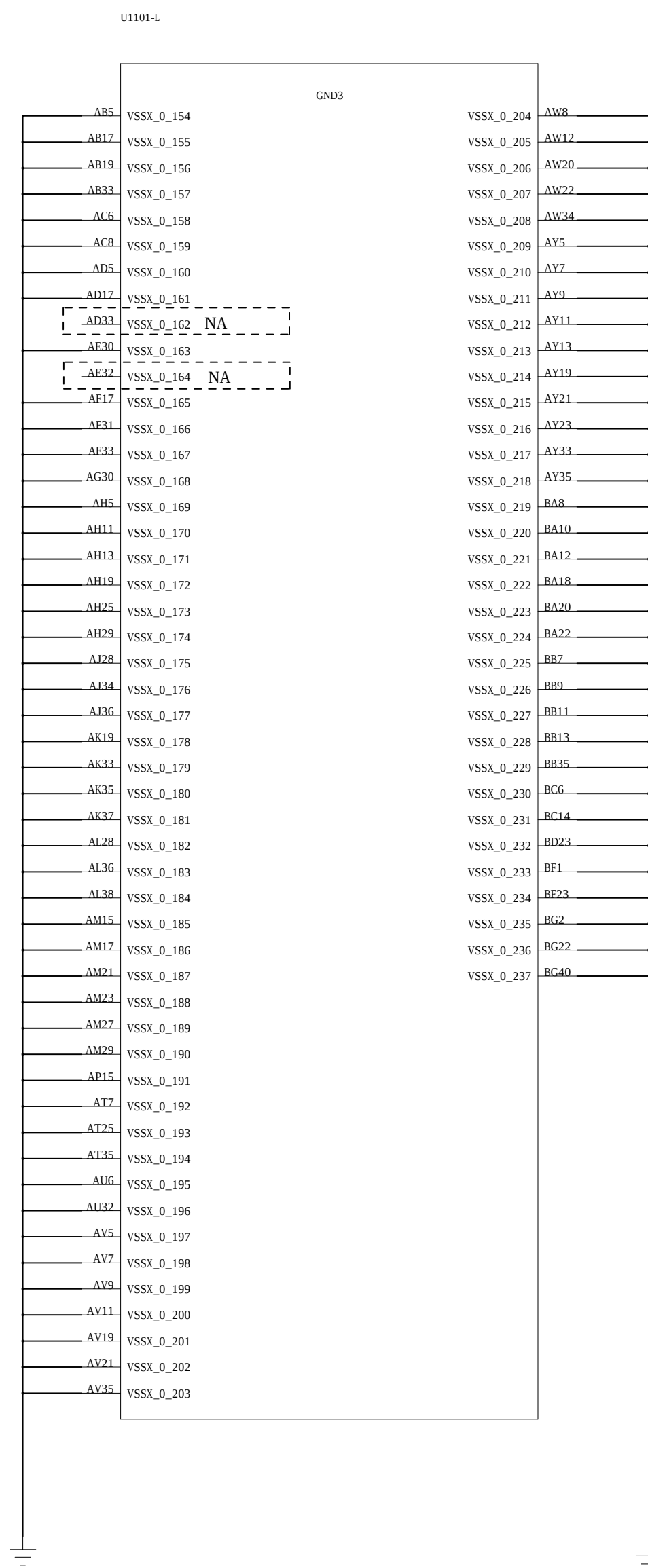


Power for secure digital calibration delay circuits

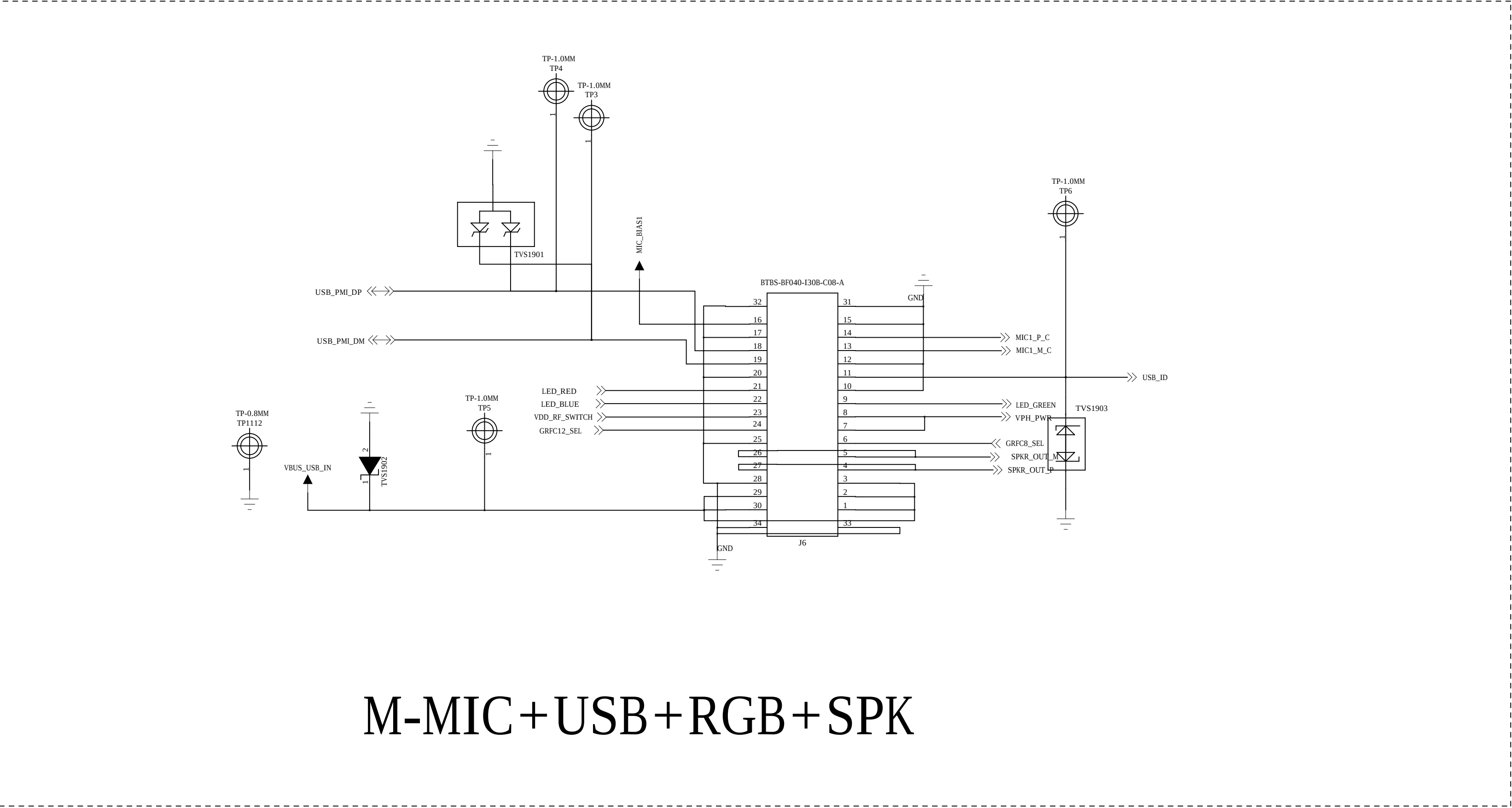
MSM8937 POWER2

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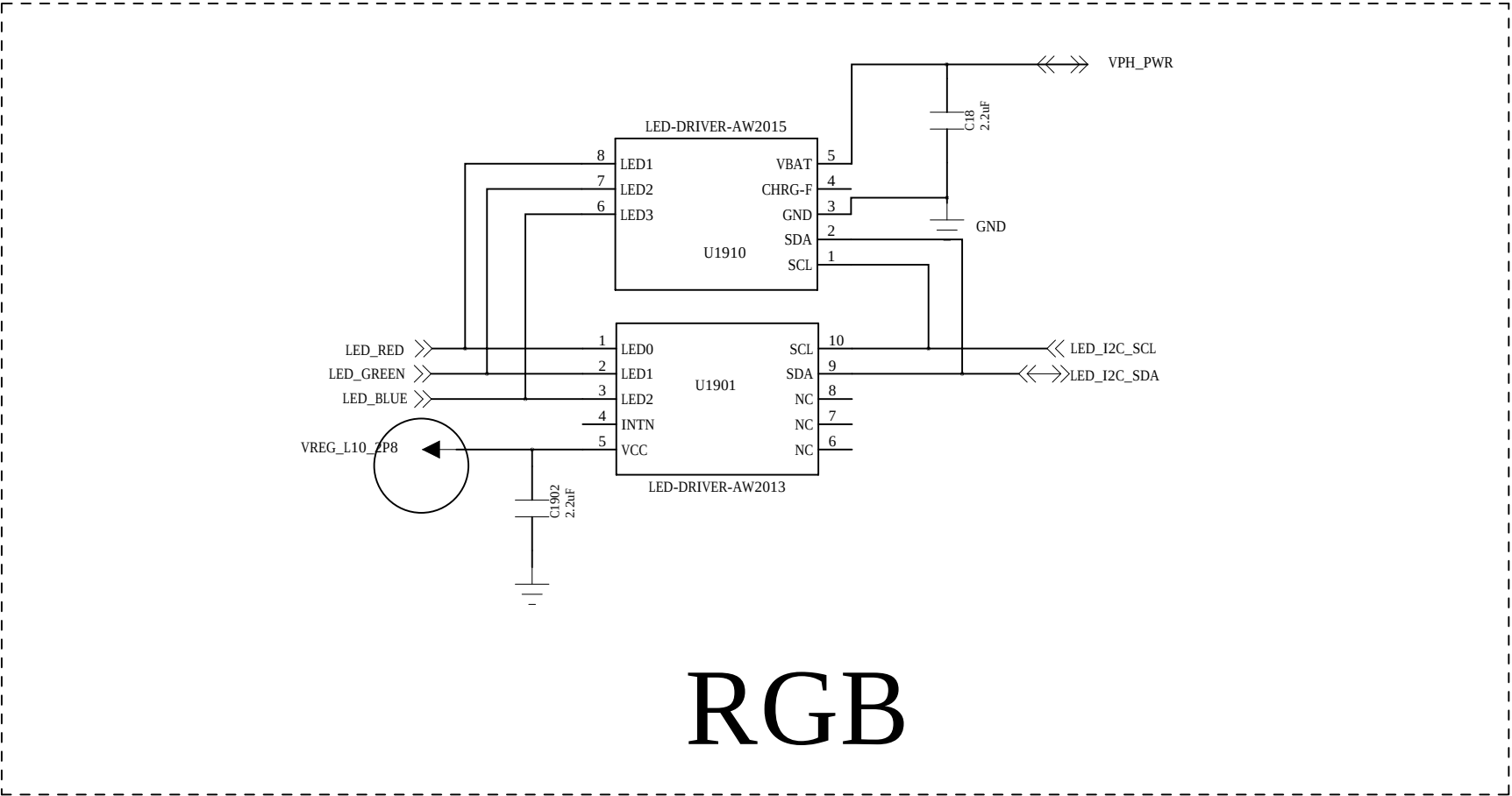
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MSM8937 GND



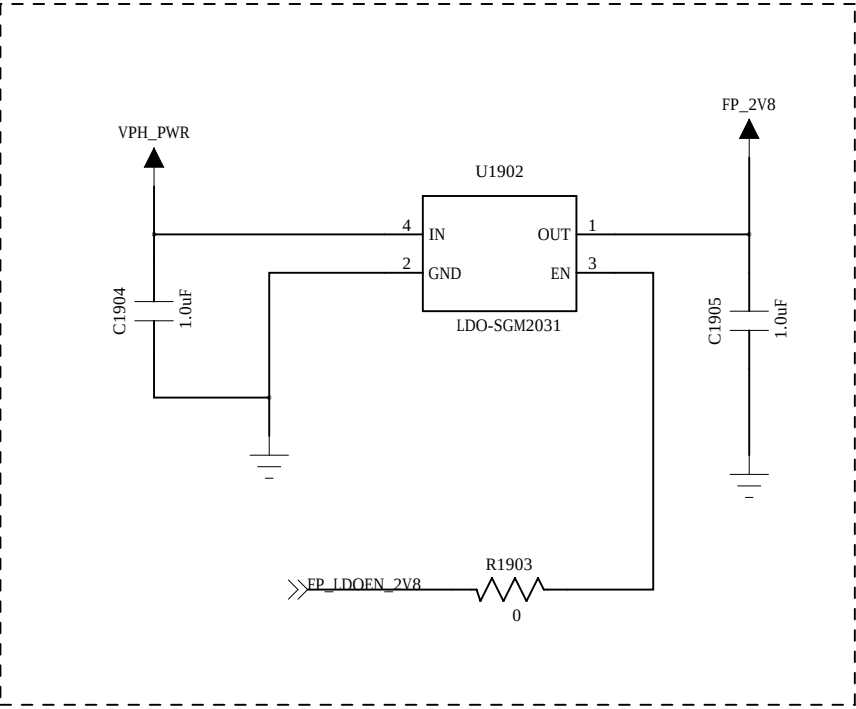
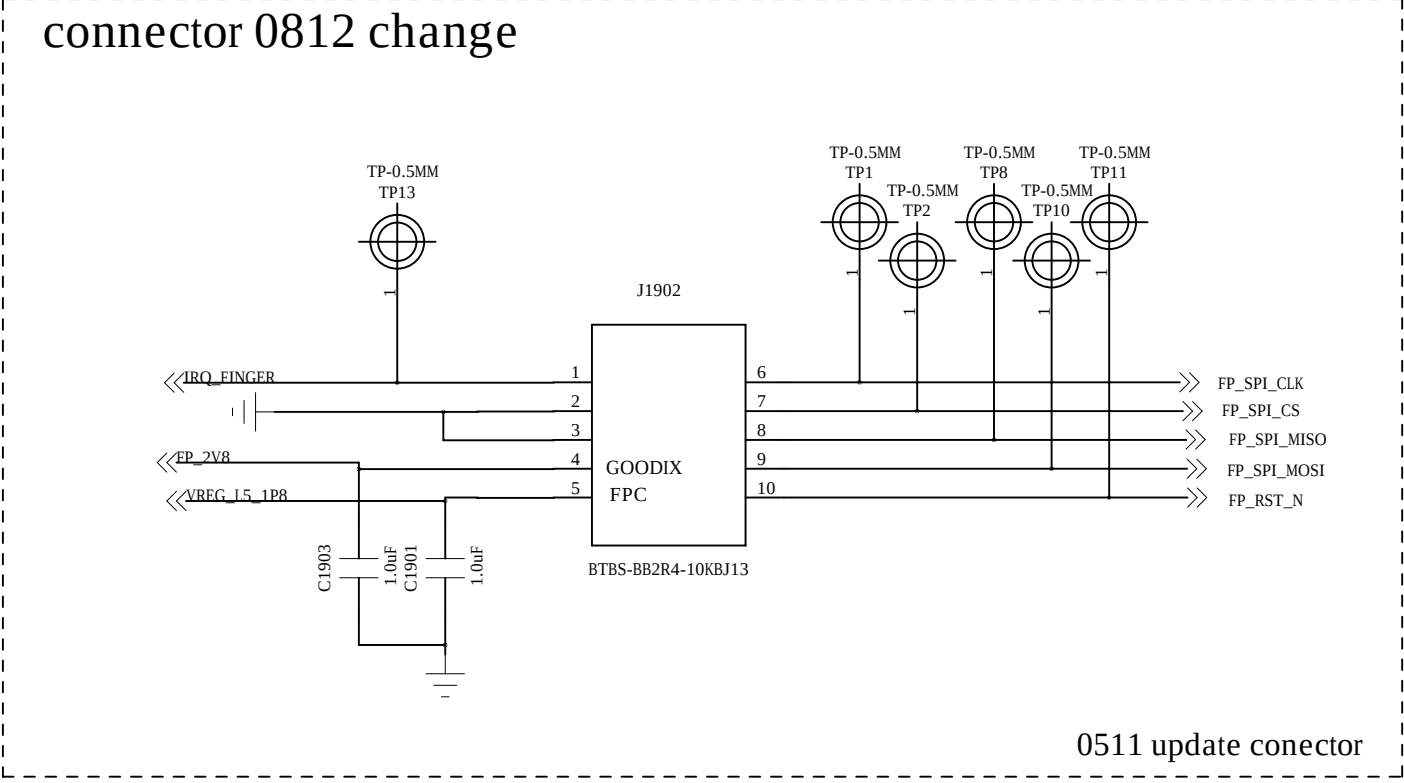
M-MIC+USB+RGB+SPK



RGB

I2C Address	
Write	8B0x8A
Read	8B0x8B

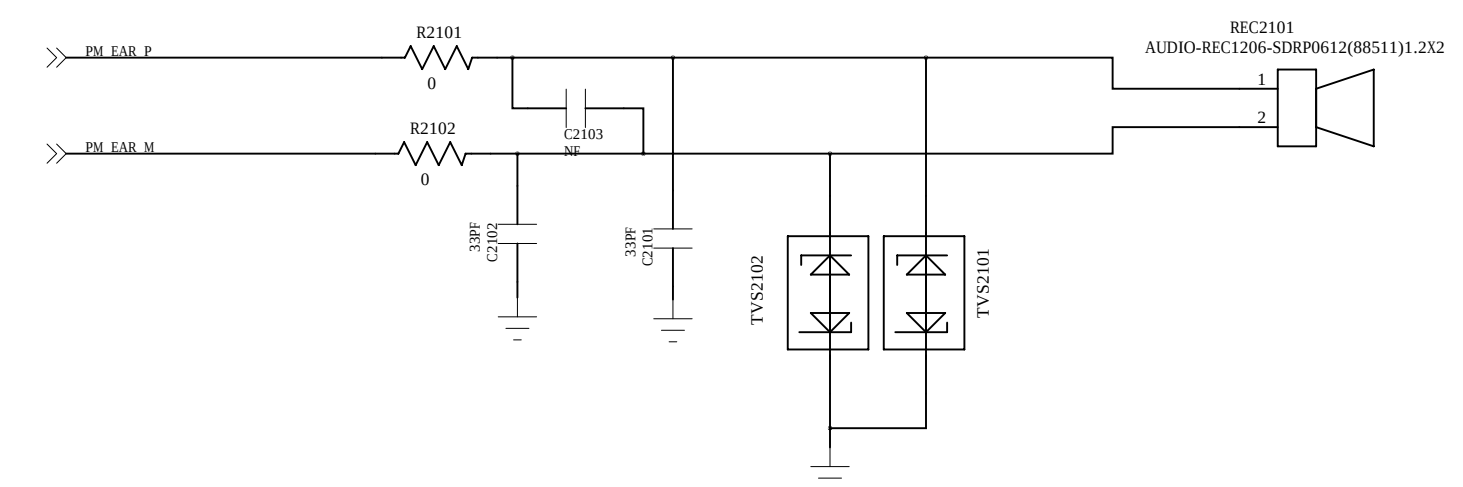
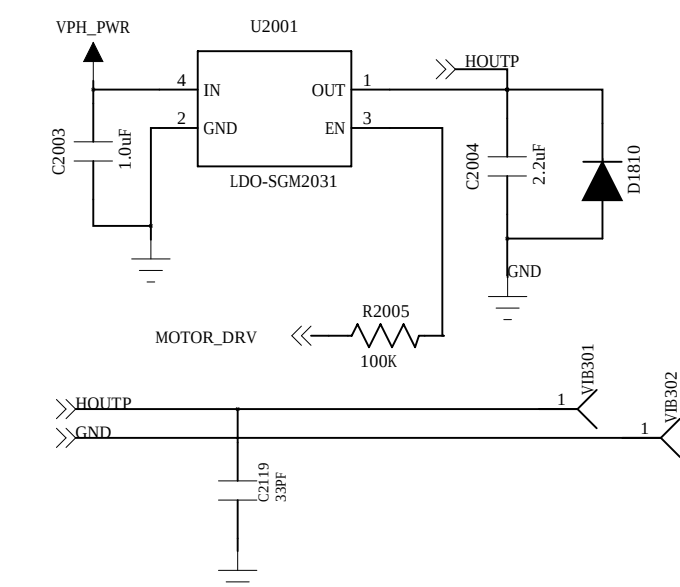
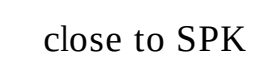
Power on sequence: can be applied in any order

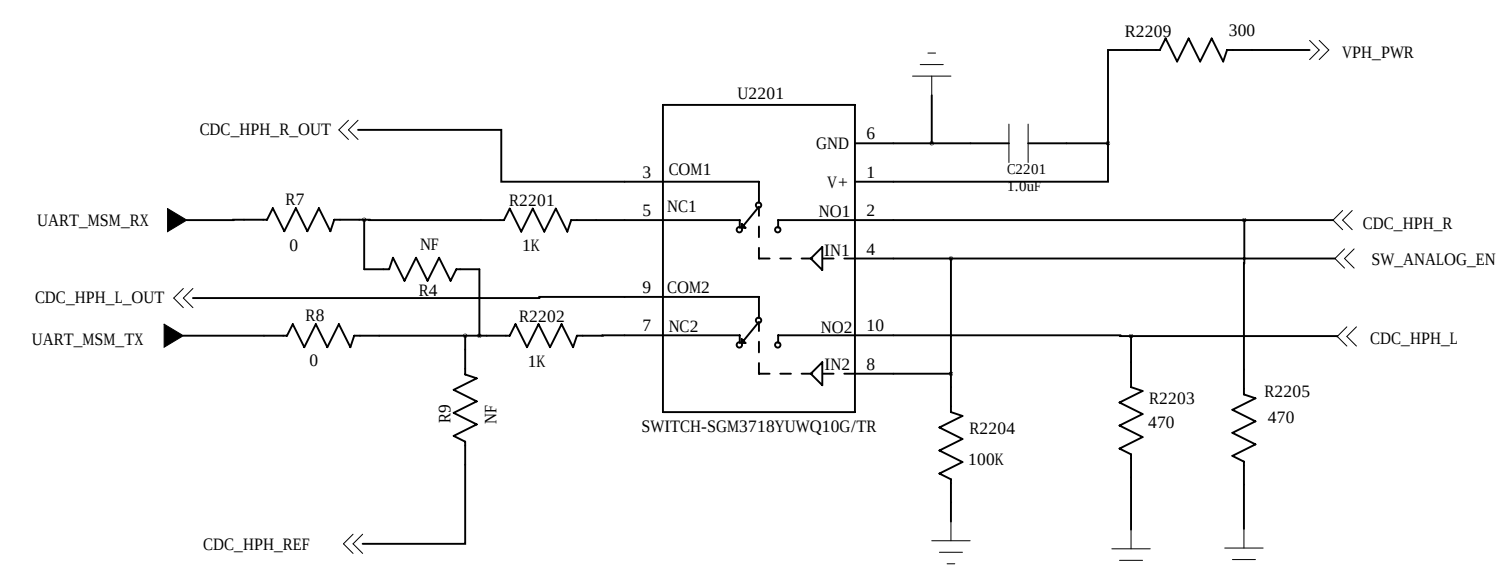


USB/SUB BOARD CONNECTOR

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AUDIO JACK

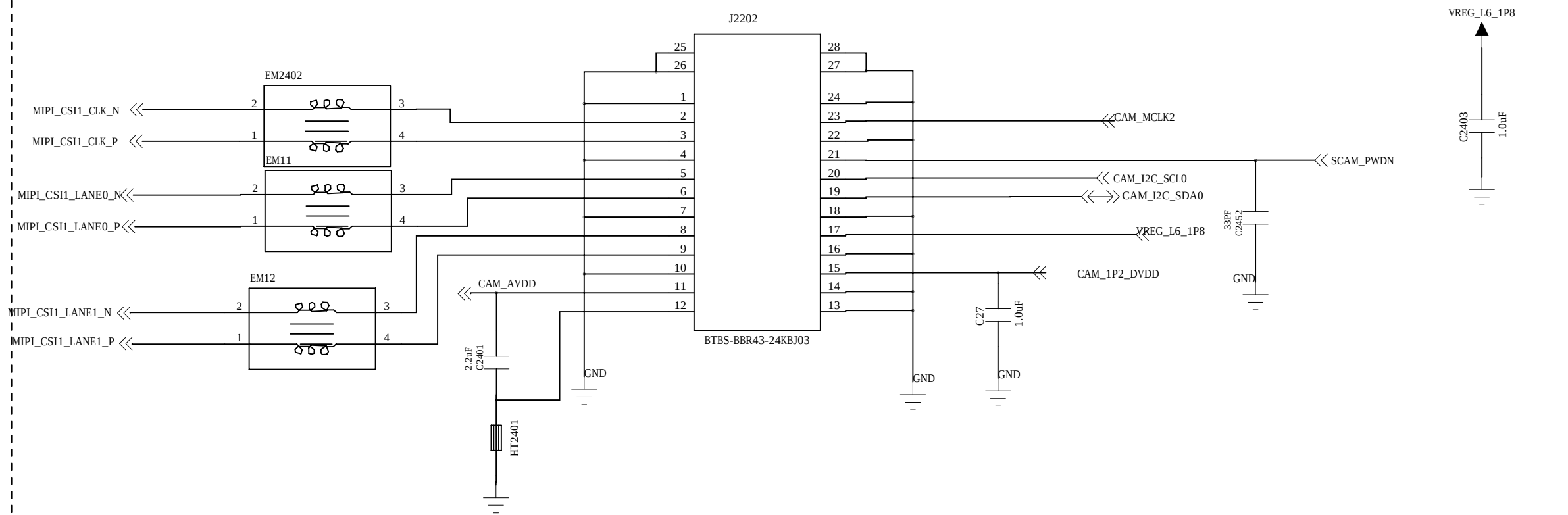


Title		
Sheet		
Size	Name	Rev
Date:	Sheet	of

1	DGND	
2	MCN	
3	MCP	
4	DGND	
5	MDNO	
6	MDPO	
7	DGND	
8	MDN1	
9	MDP1	
10	DGND	
11	AVDD	2.8V
12	AGND	
13	DGND	
14	DGND	
15	DVDD	1.2V
16	DGND	
17	DOVDD	1.8V
18	DGND	
19	SDA	
20	SCL	
21	RESET	
22	DGND	
23	MCLK	
24	DGND	

FRONT_camera

0511 update conector

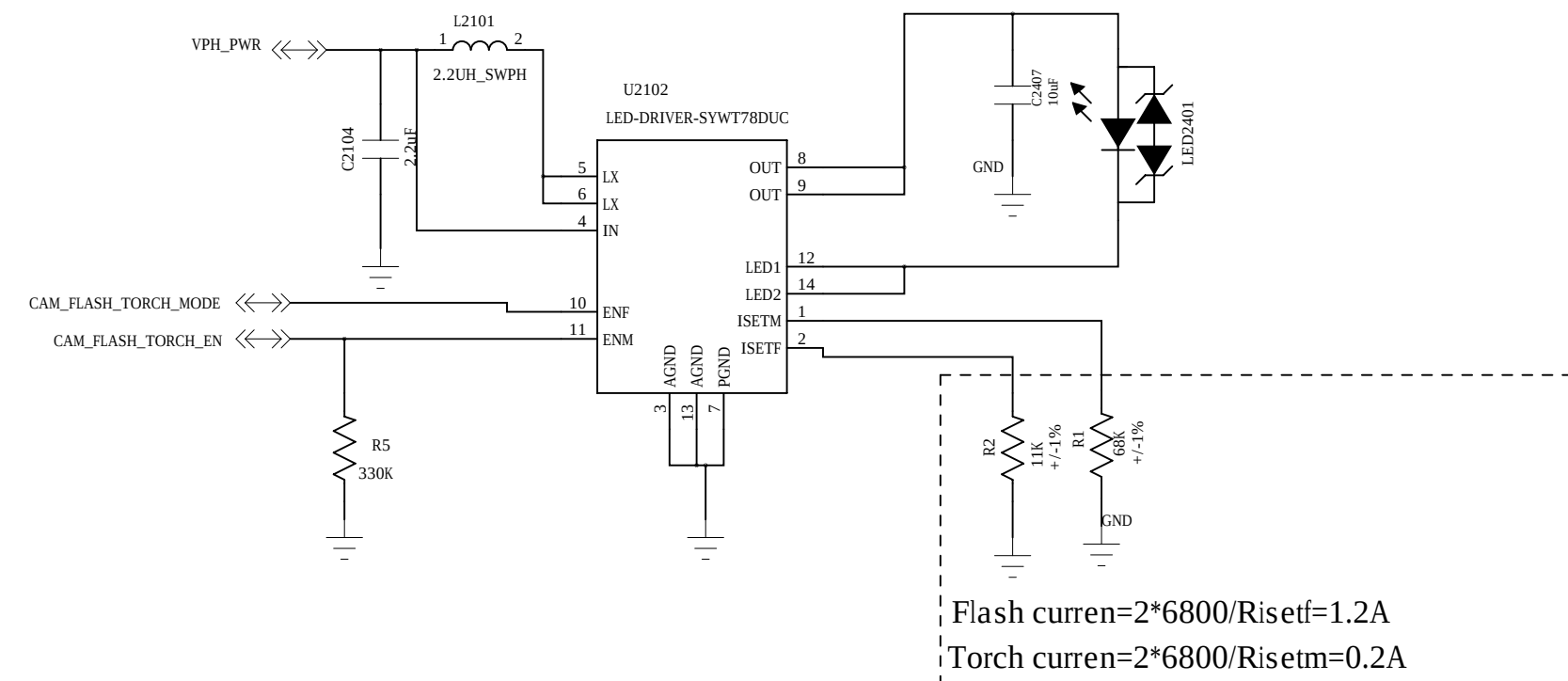
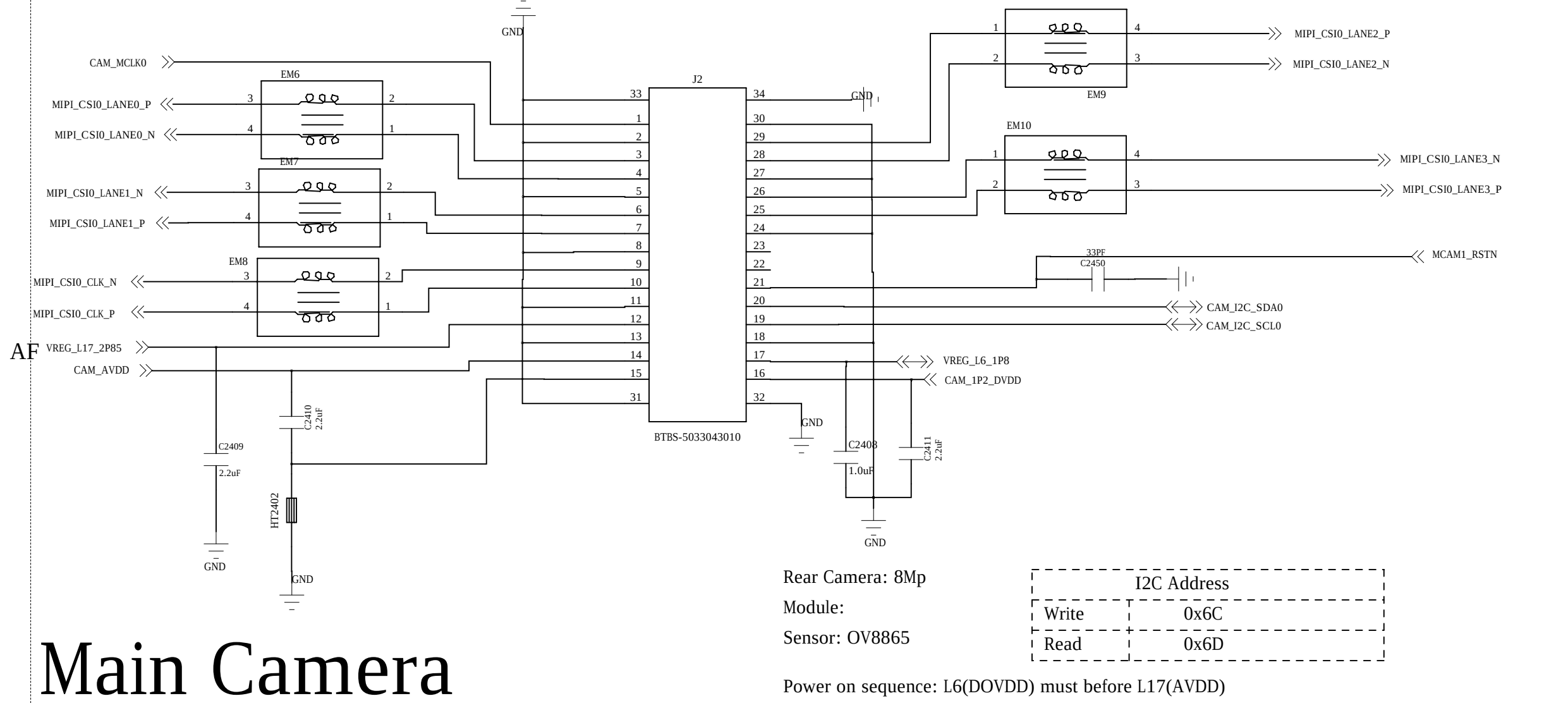


Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

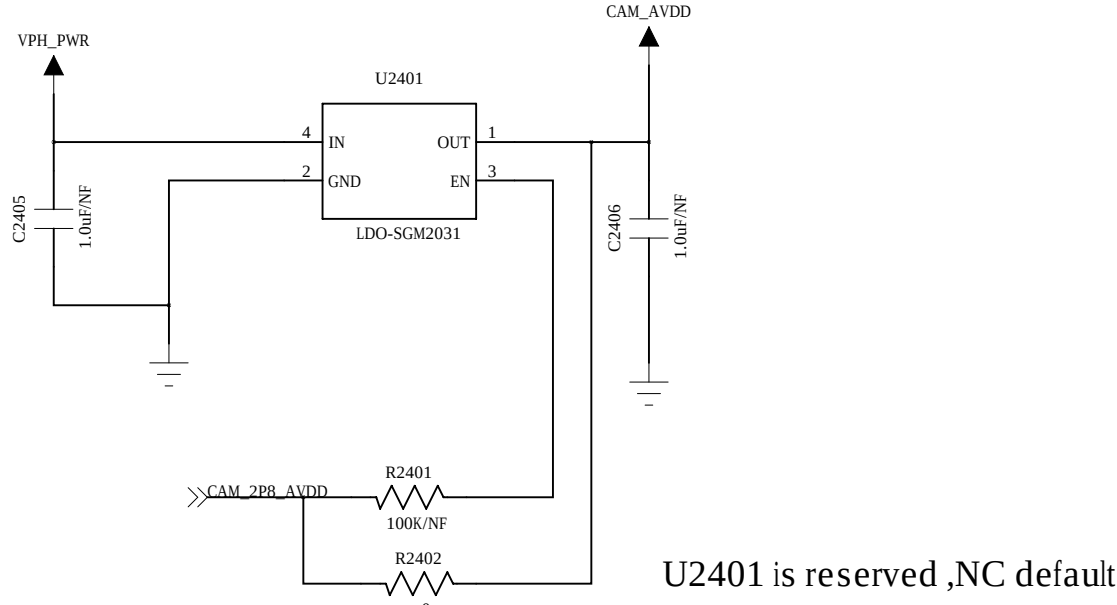
PIN	
1	MCLK
2	DGND
3	MDPO
4	MDNO
5	DGND
6	MDN1
7	MDP1
8	DGND
9	MCN
10	MCP
11	DGND
12	AF_Power
13	AF_GND
14	AVDD
15	AGND
16	DVDD
17	DOVDD
18	DGND
19	SIO_C
20	SIO_D
21	RESET
22	VPP
23	AF_EN
24	DGND
25	MDP3
26	MDN3
27	DGND
28	MDN2
29	MDP2
30	DGND

Main Camera

0711 update conector



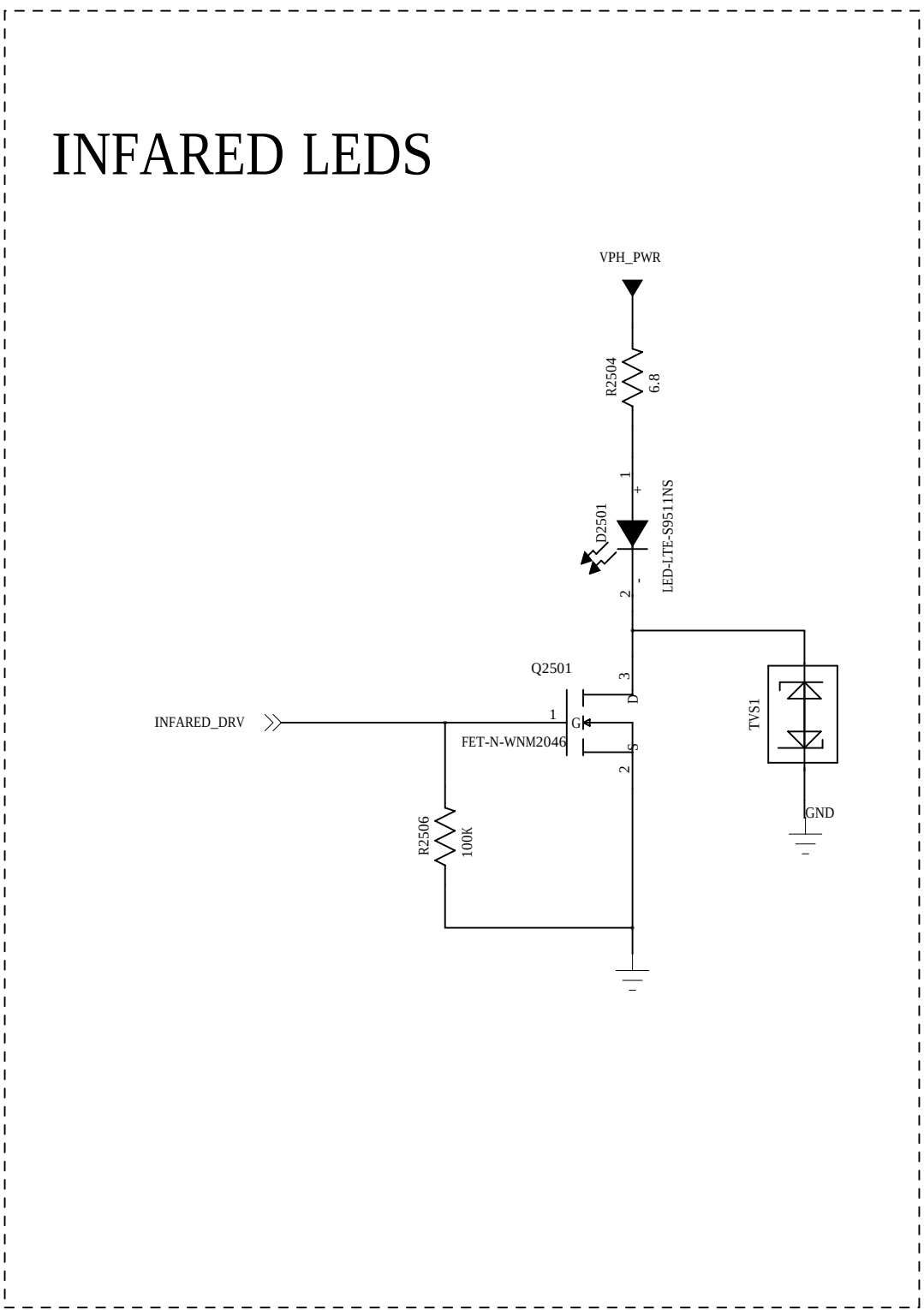
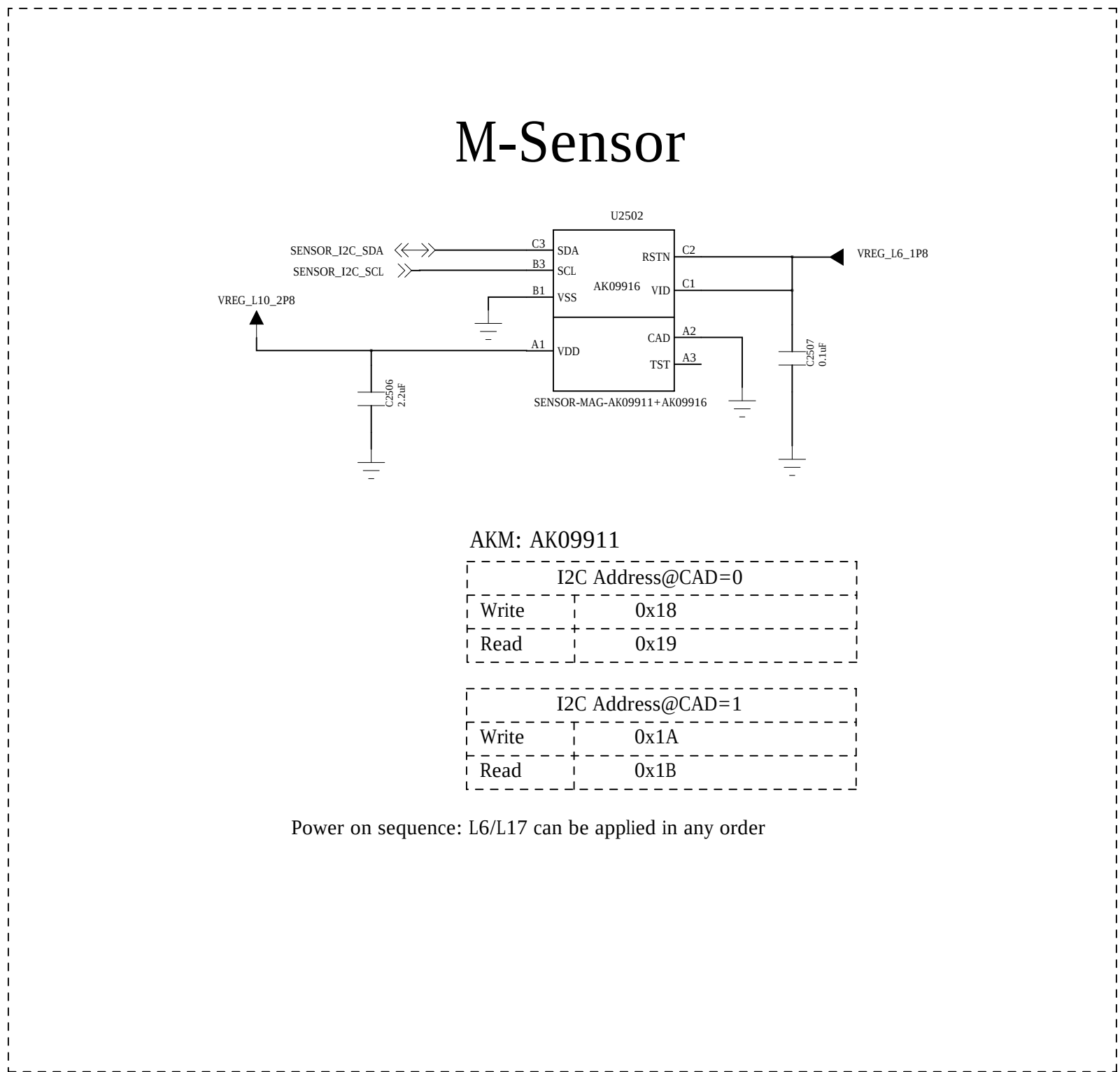
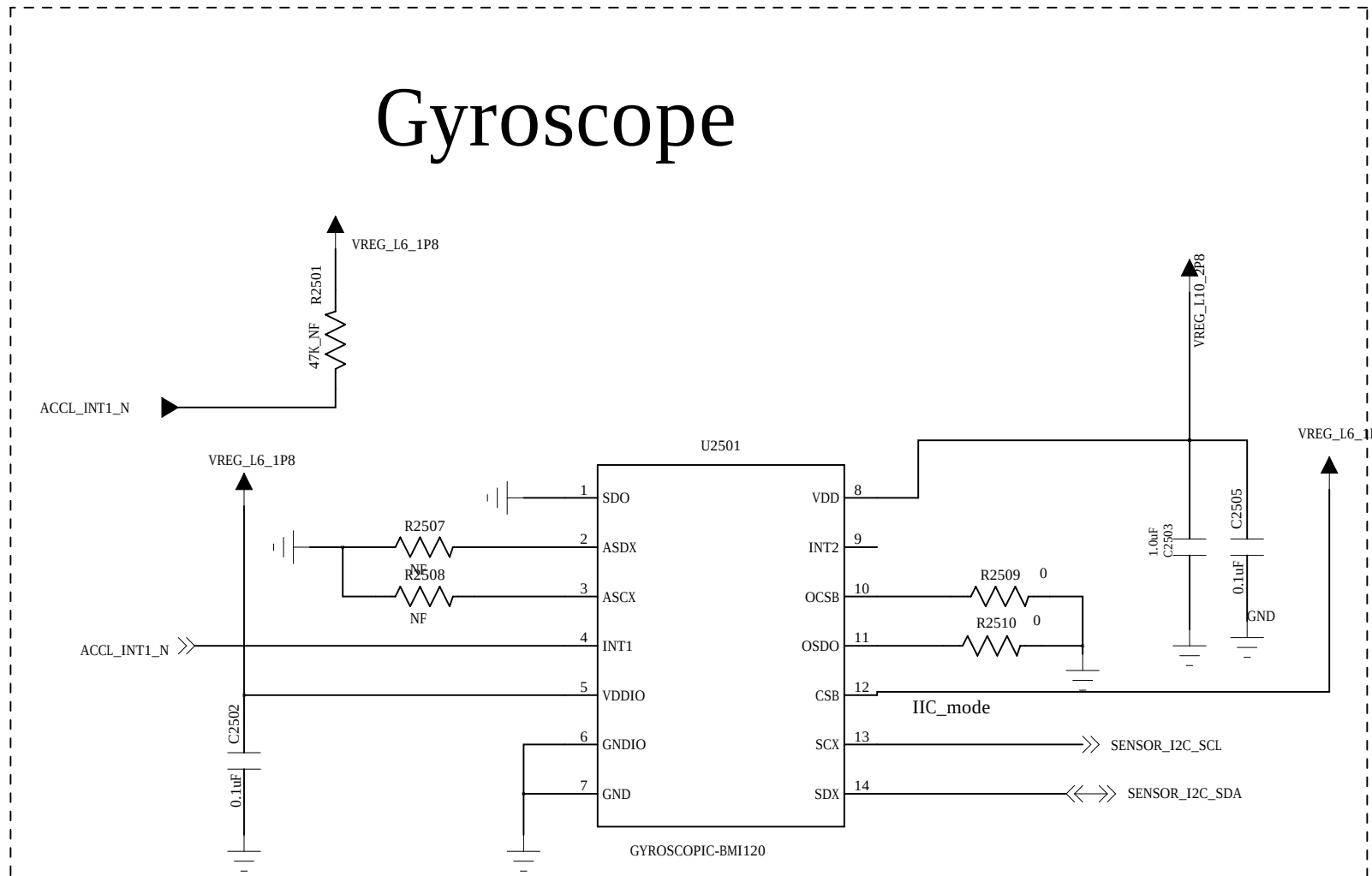
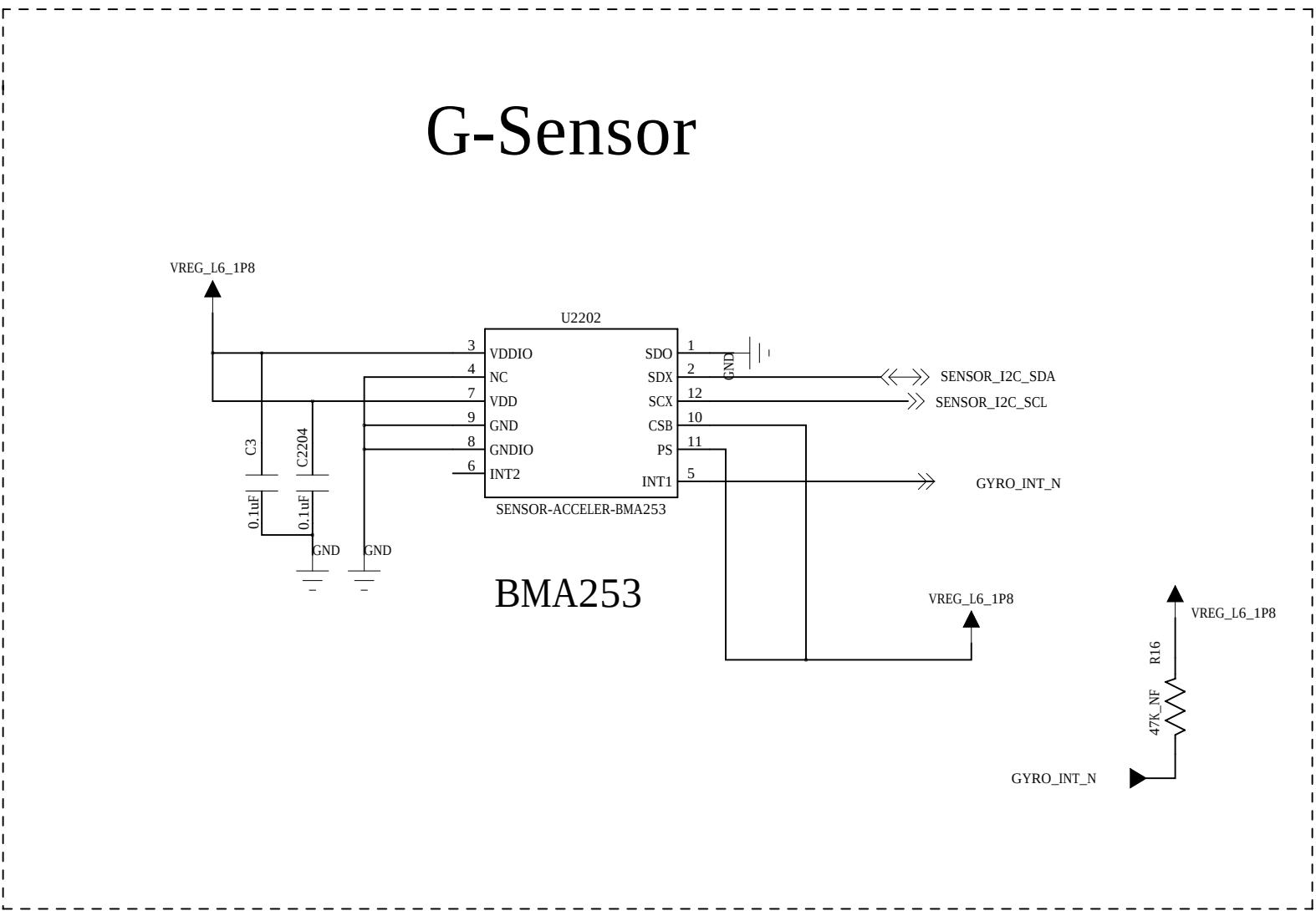
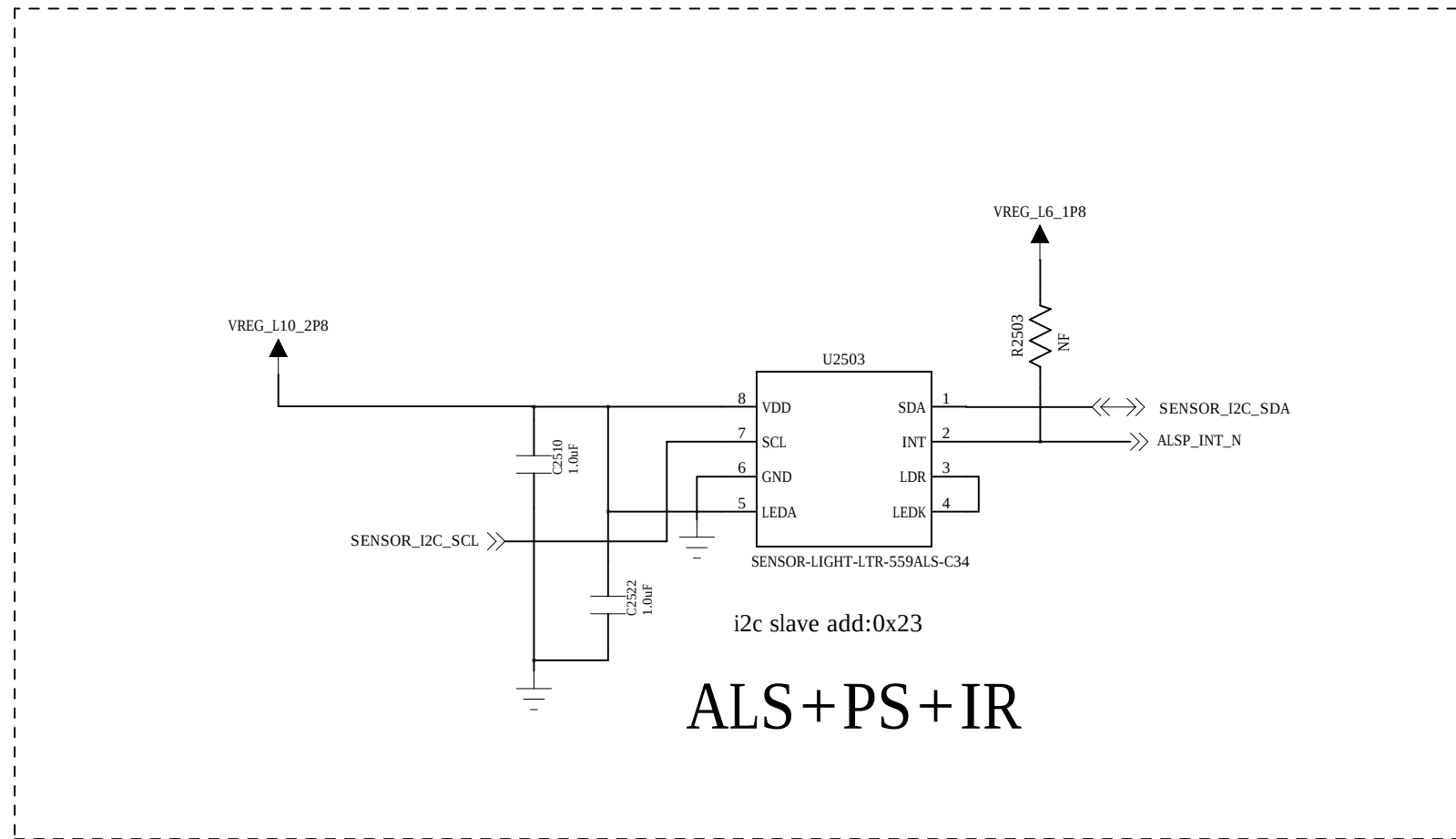
FLASH LED DRIVER



Main/Slave Camera and Flash

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BMI160 SD0=0 7b1101000 0x68
SD0=1 7b1101001 0x69

LSM6DS3 SD0=0 7b1101010 0x6A
SD0=1 7b1101011 0x6B

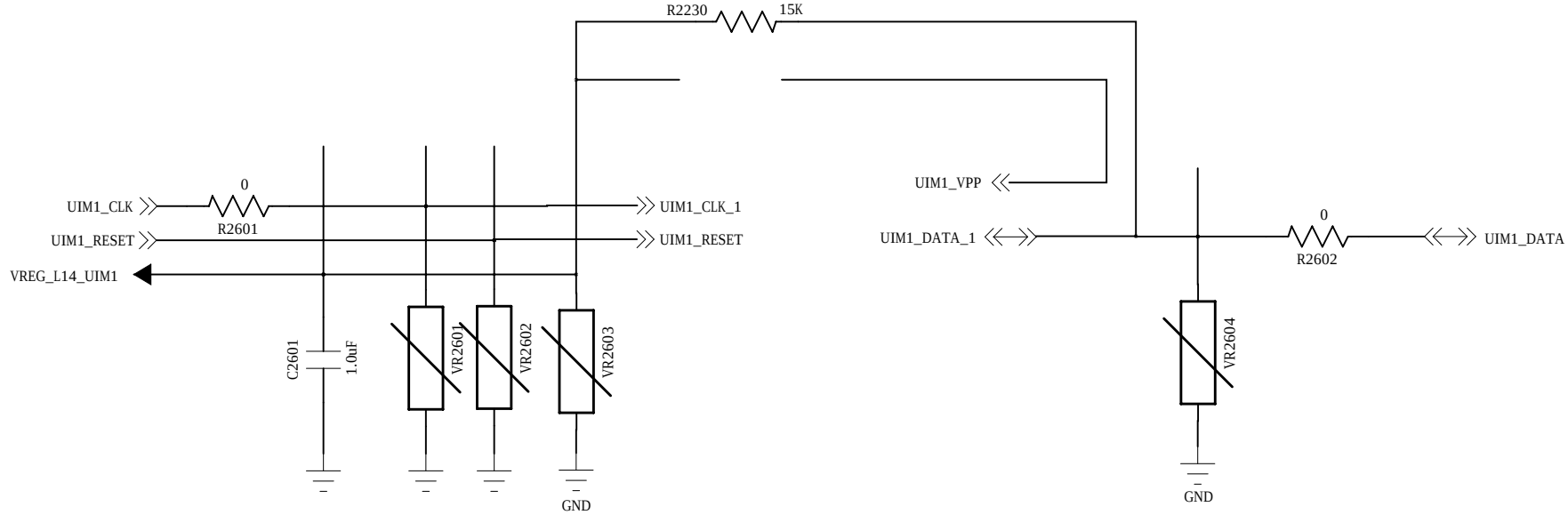
Power on sequence: L6/L17 can be applied in any order

SENSORS

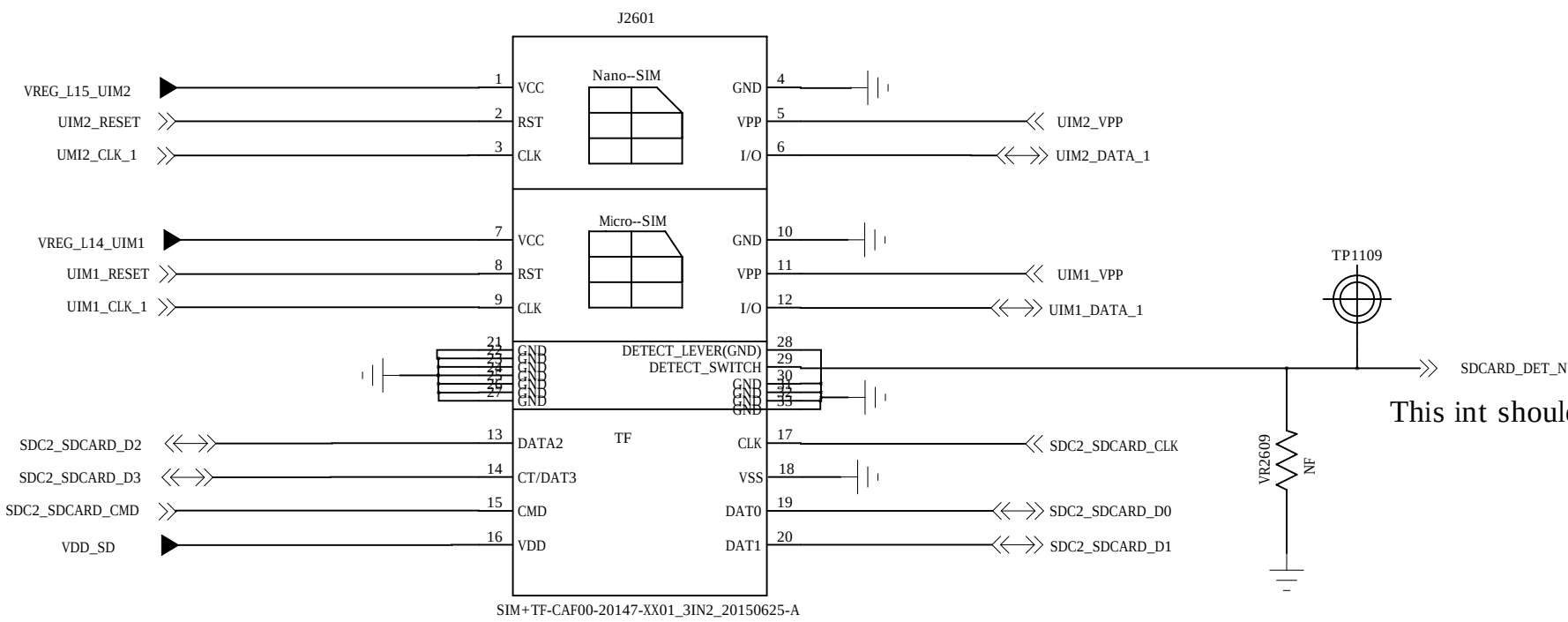
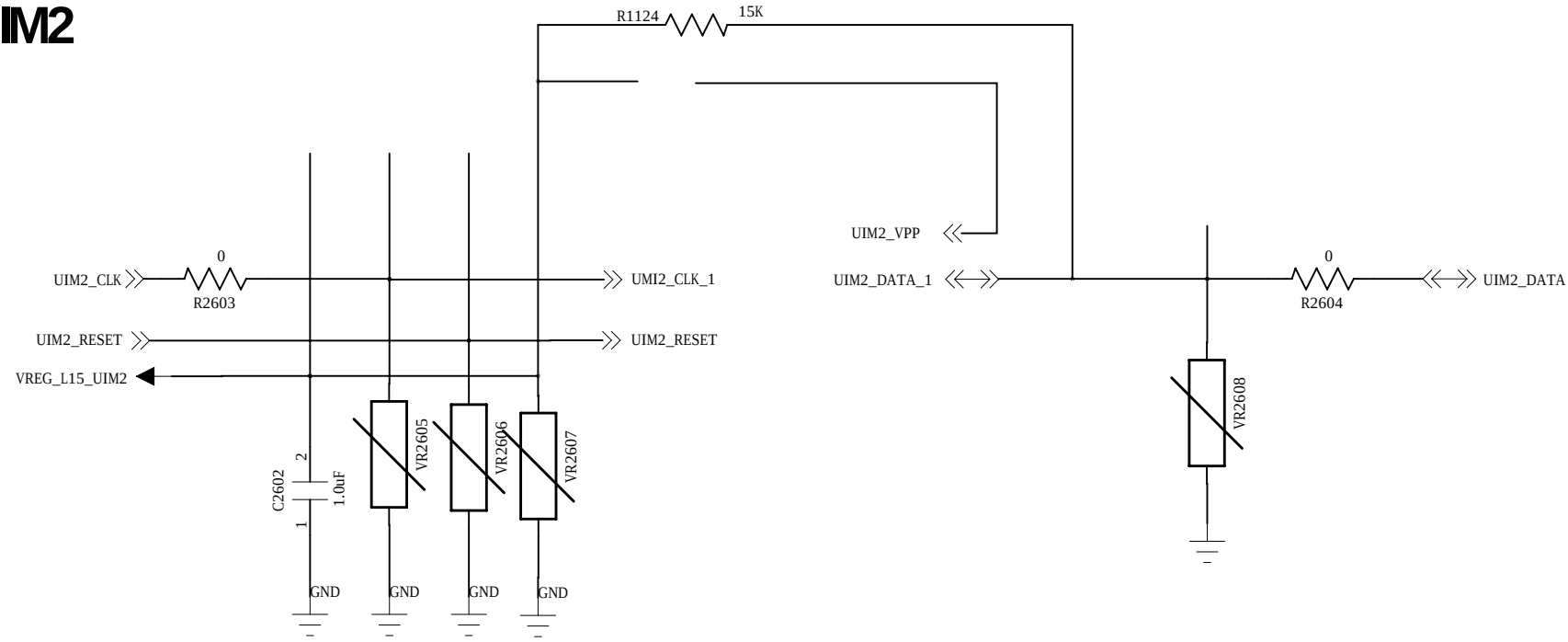
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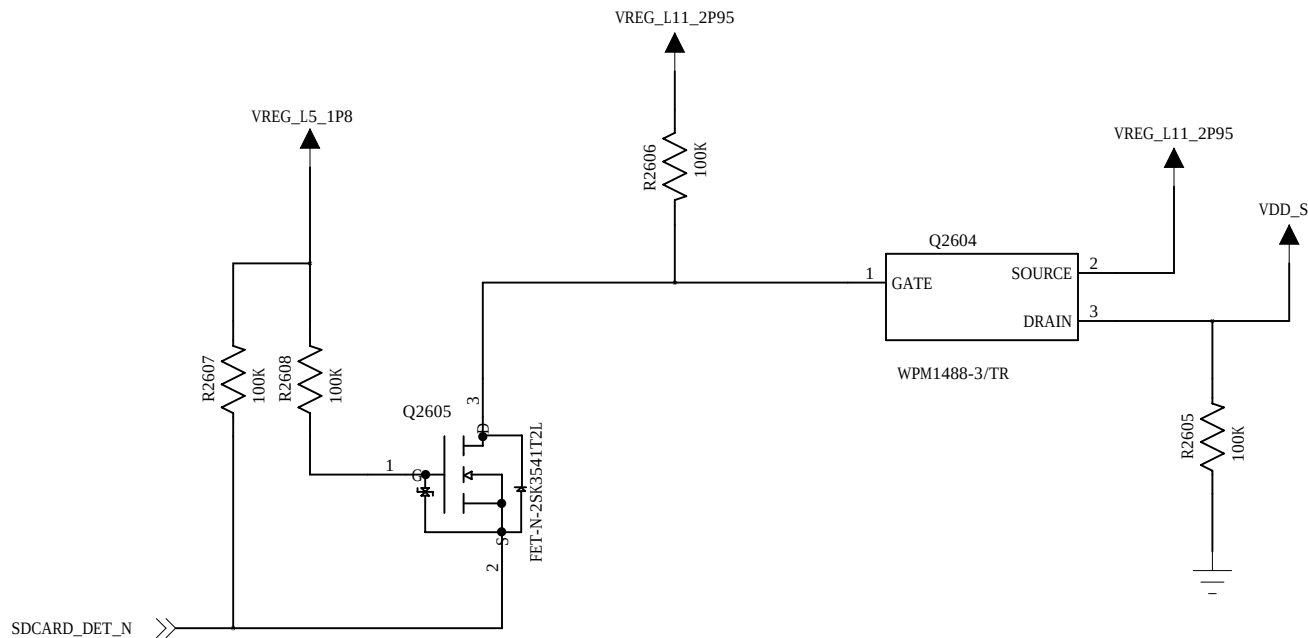
SIM1



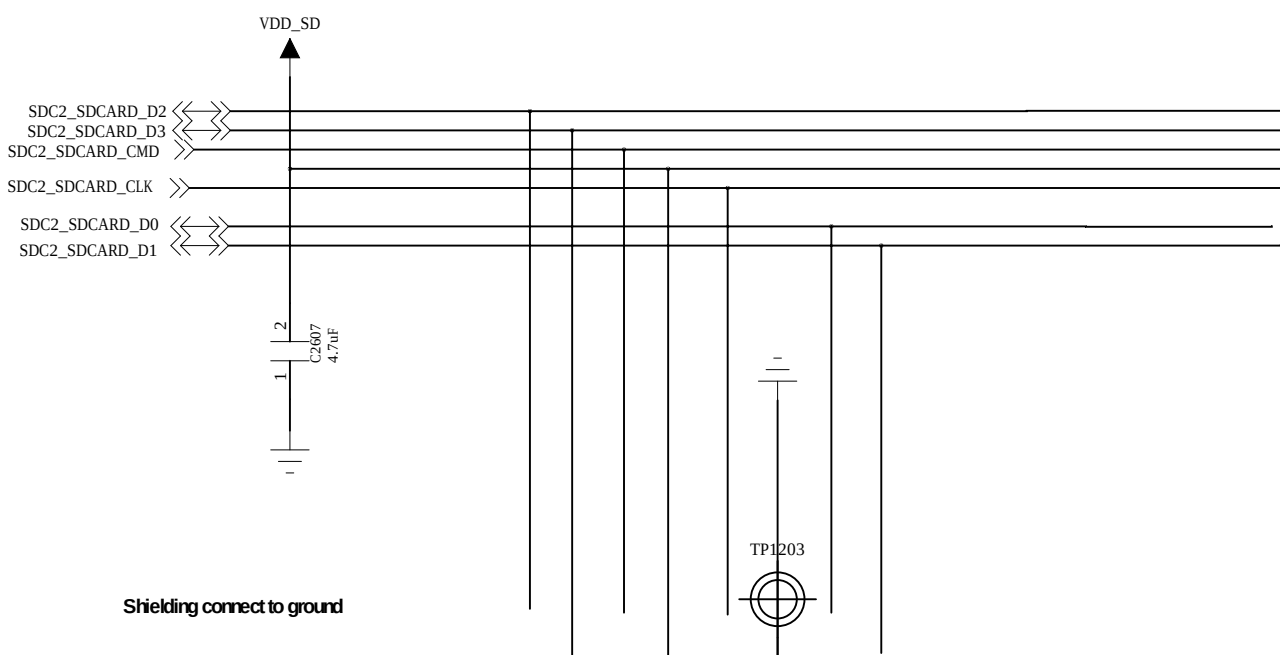
SIM2



Vout=3.0V,I-lim=1A



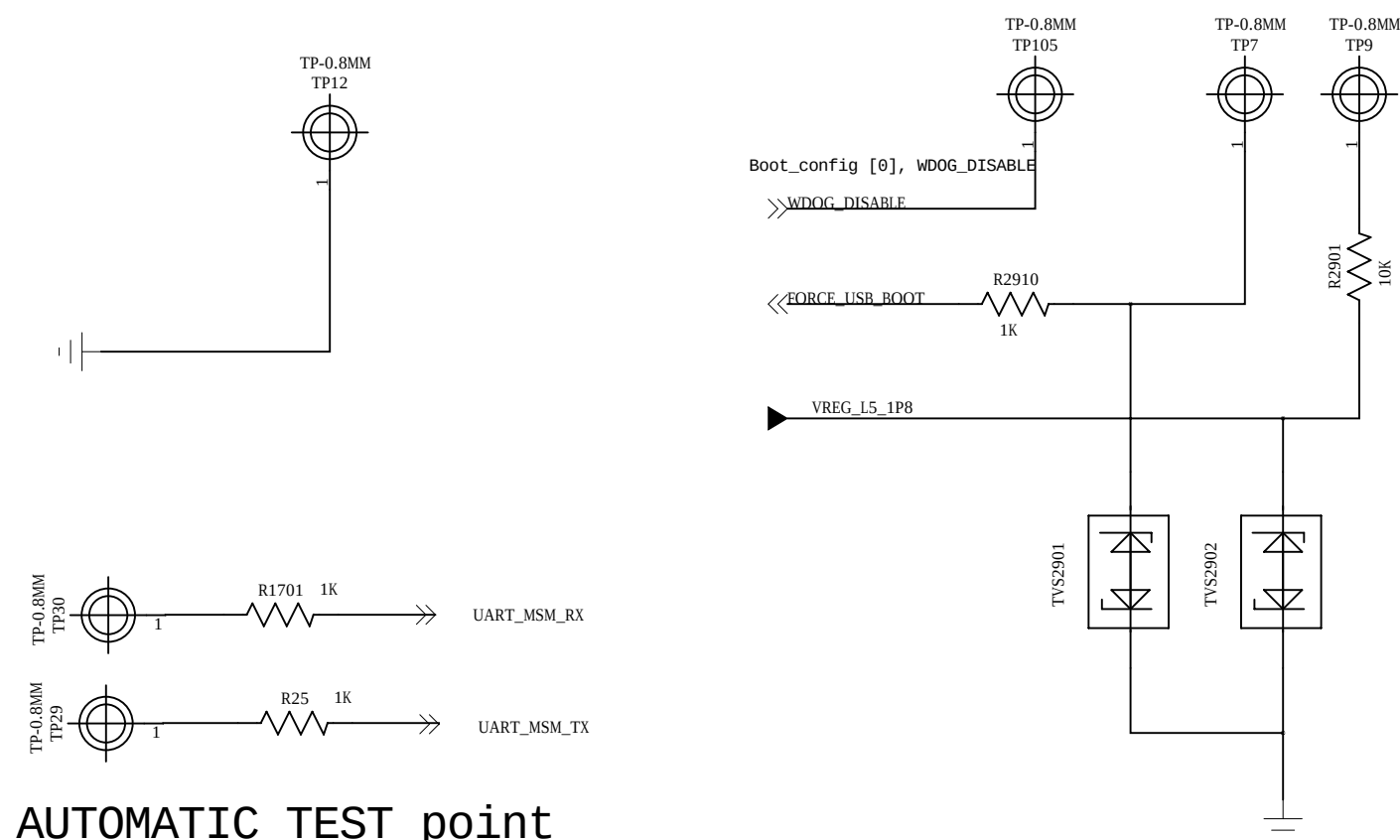
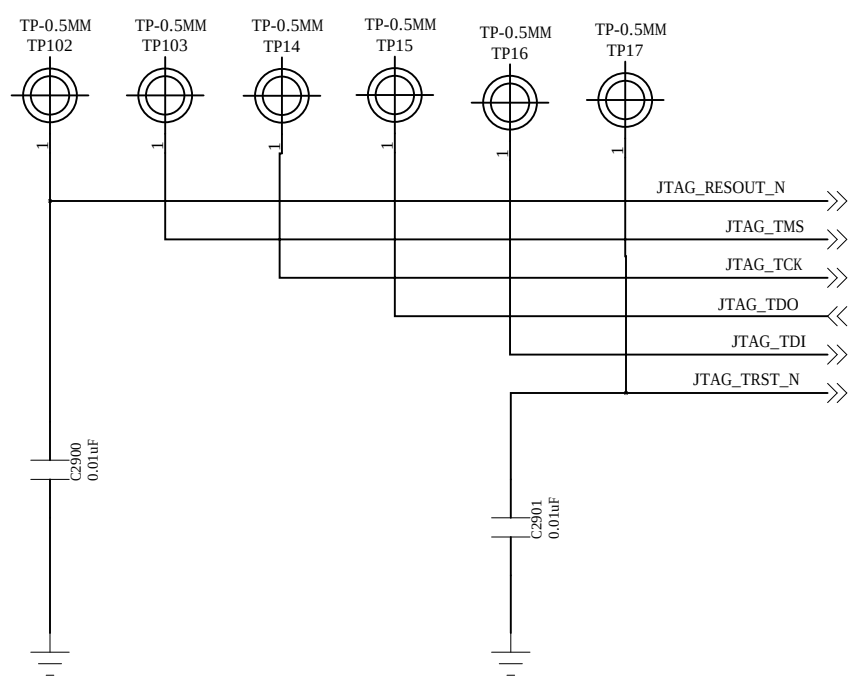
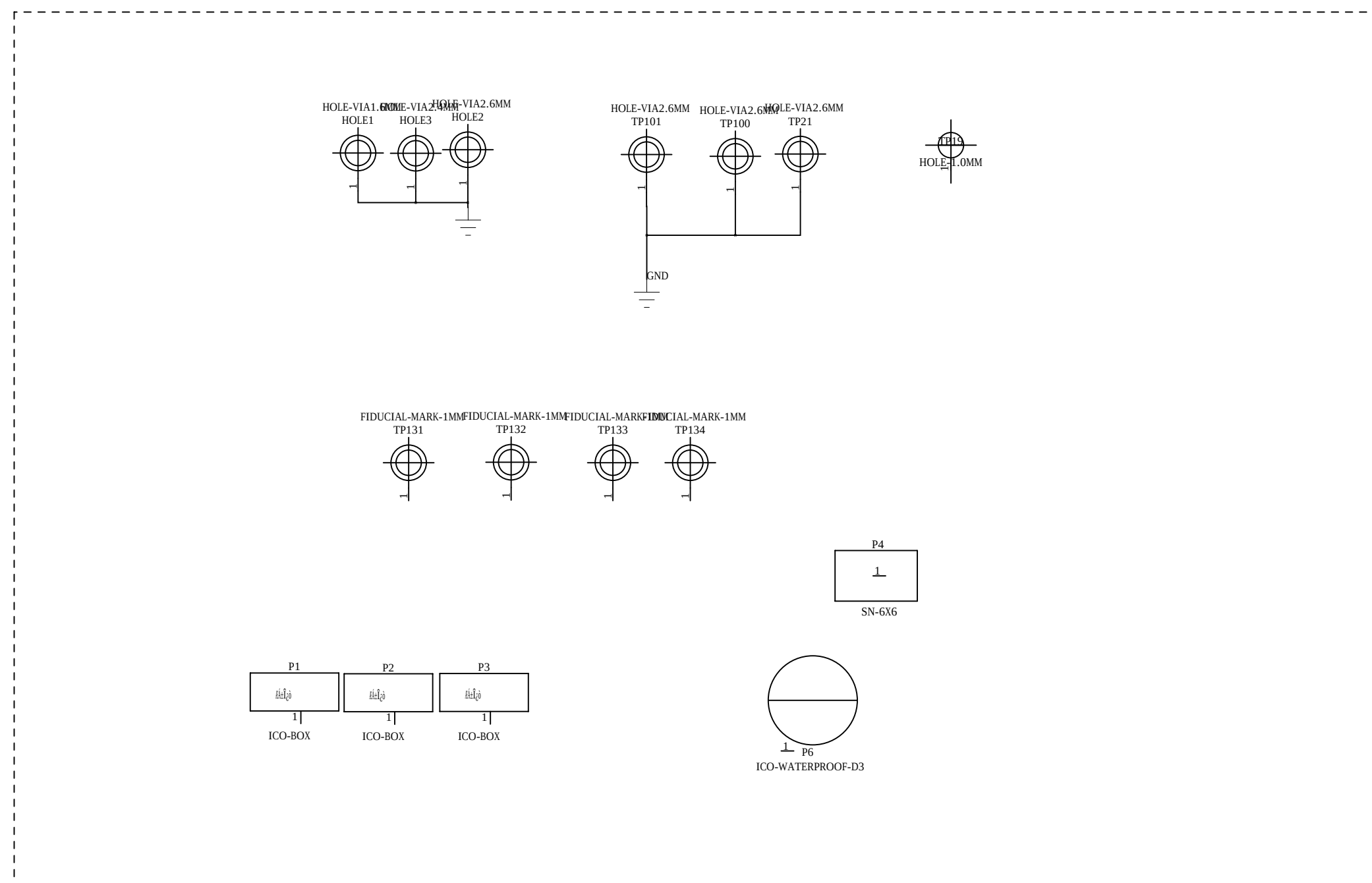
SD CARD



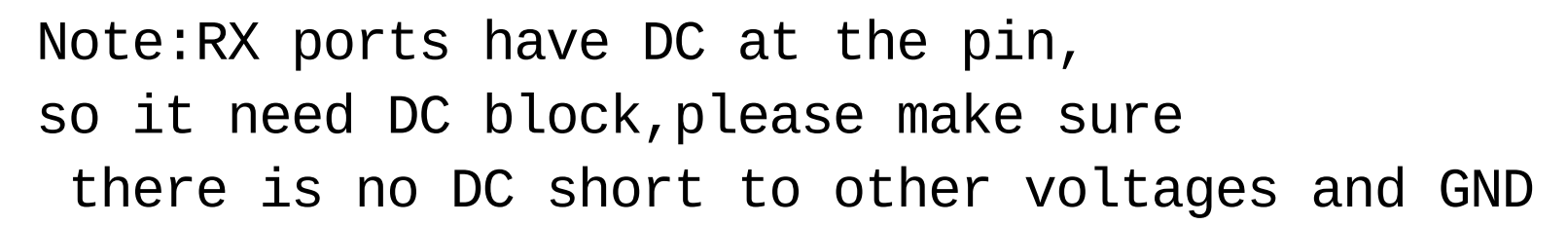
SIM/TF card

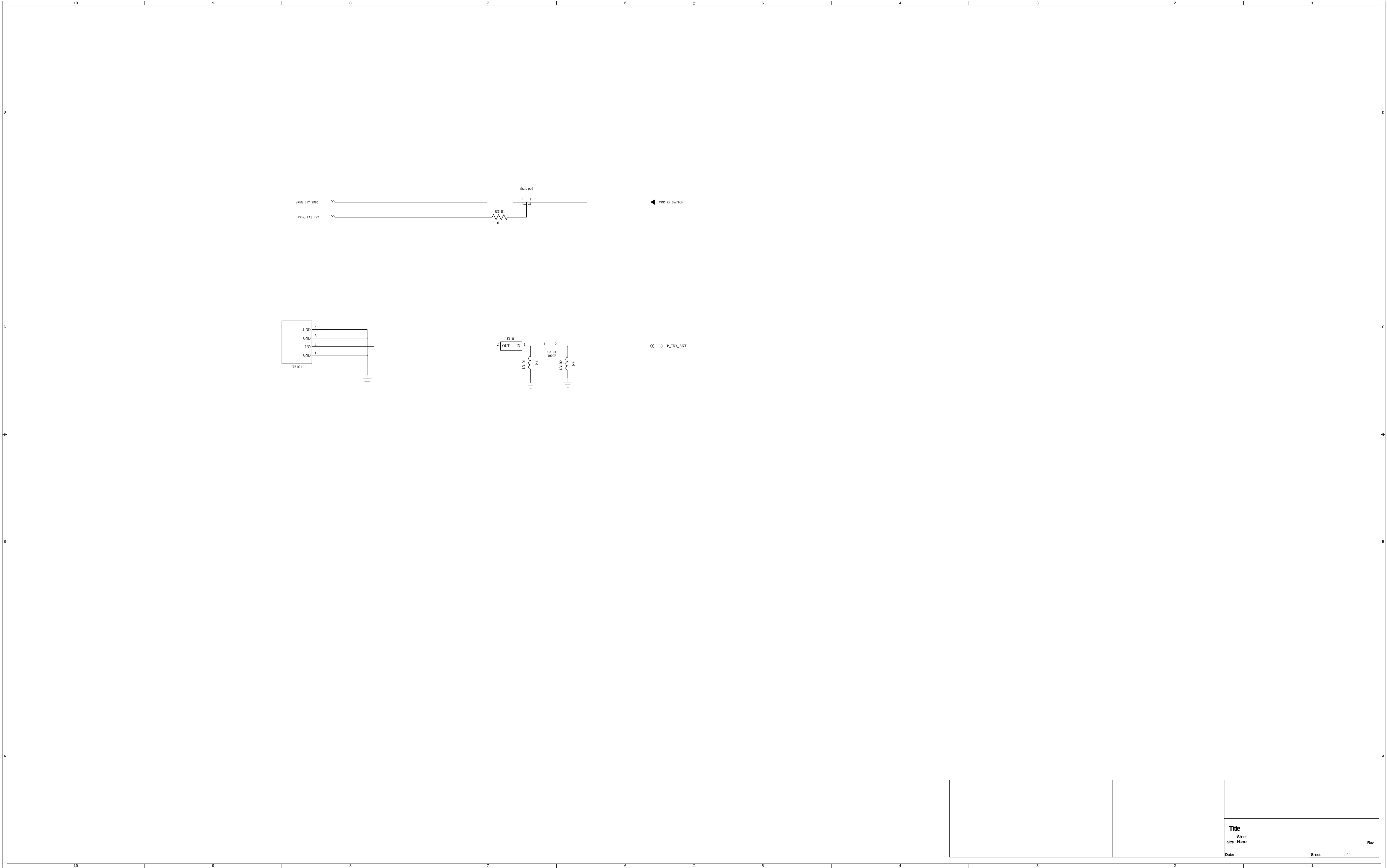
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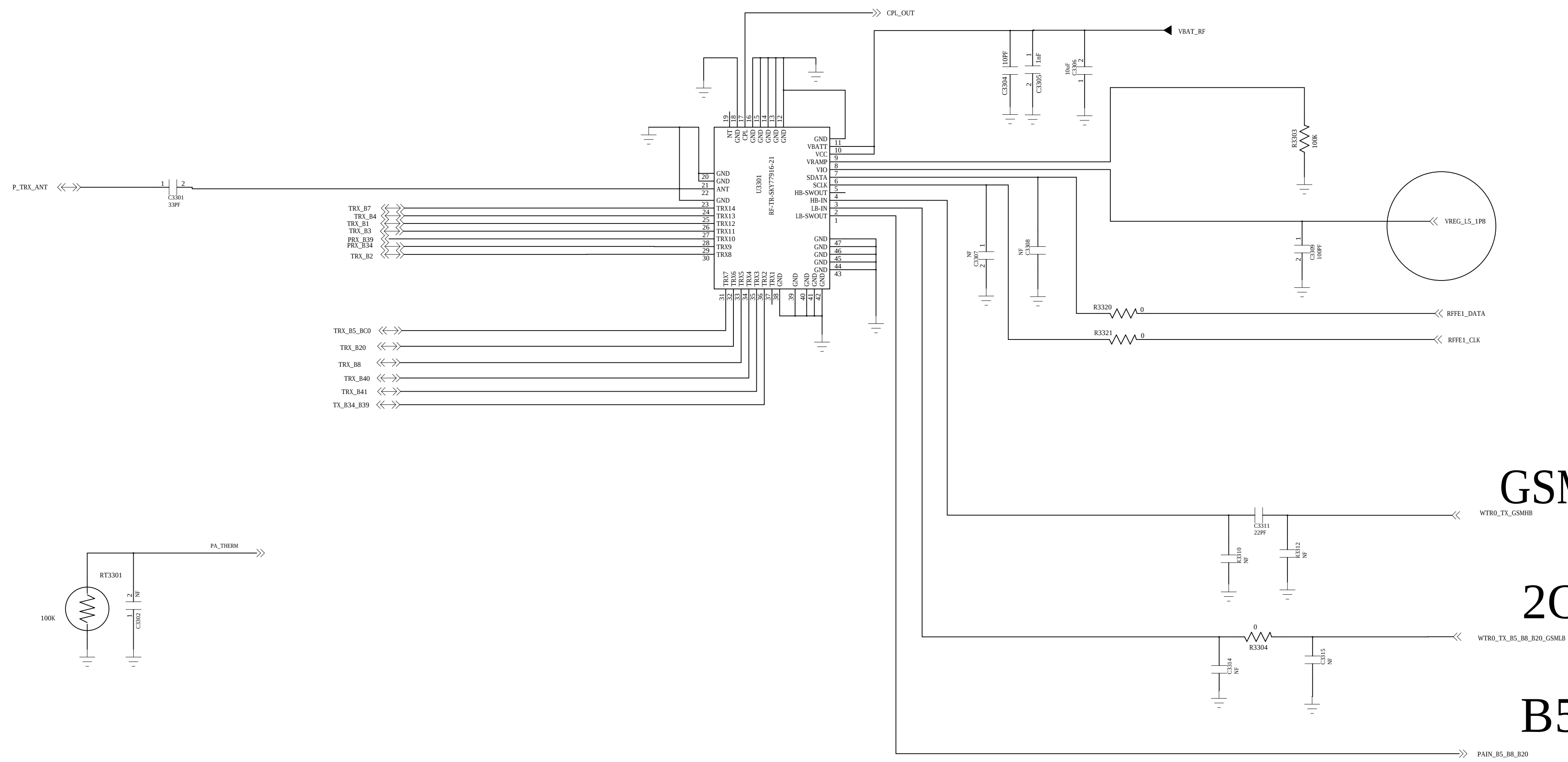


Test Point/Shields





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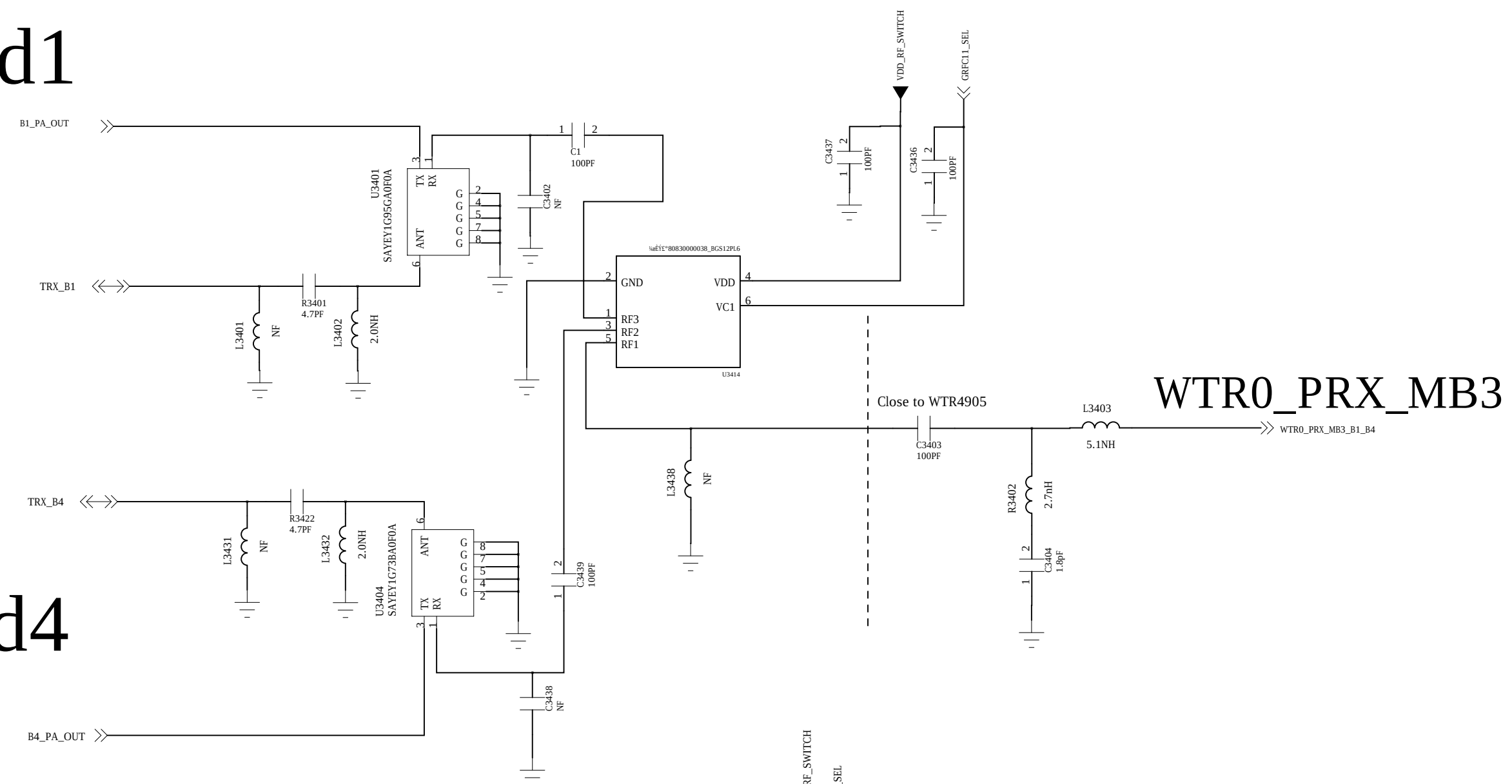
GSM_HB

2G3G4G_LB

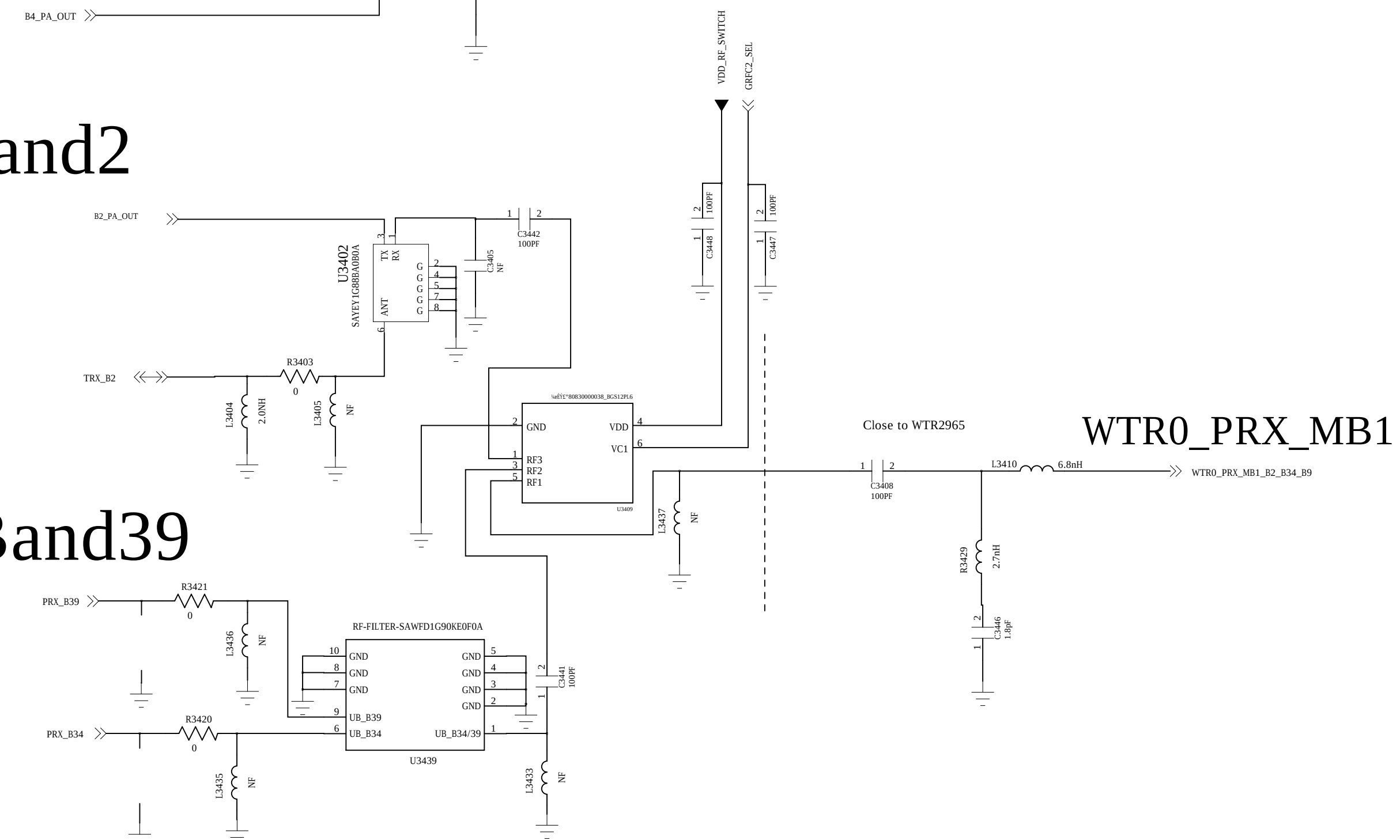
B5_B8_B20

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Sheet					
Size	Name				Rev
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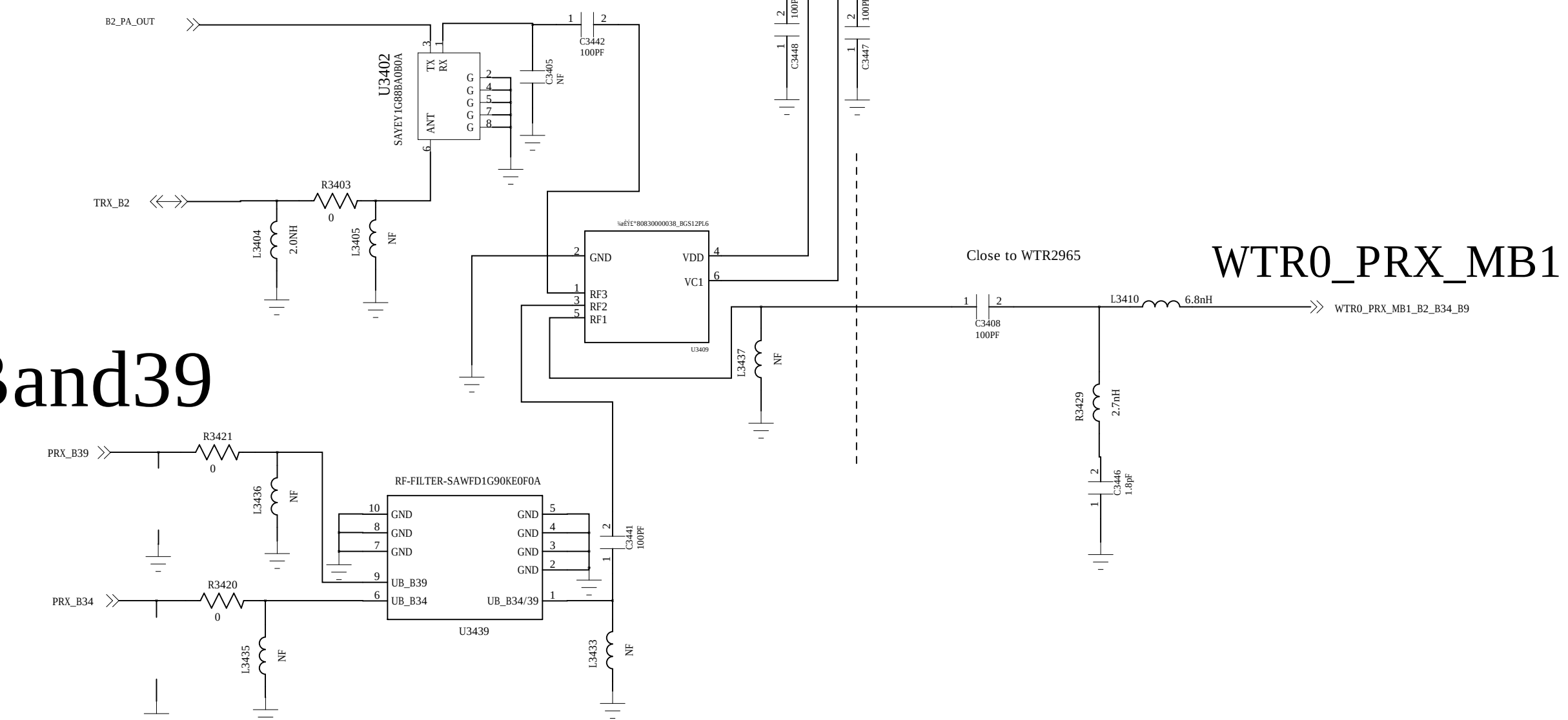
Band1



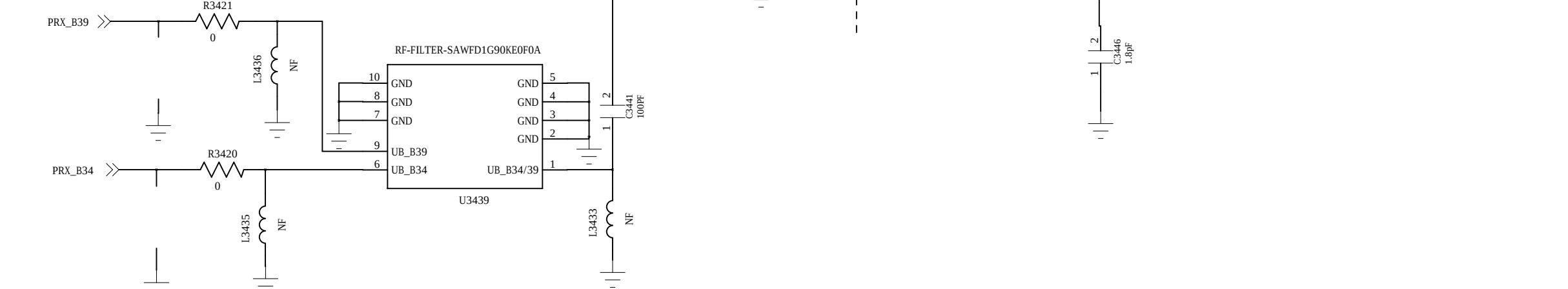
Band4



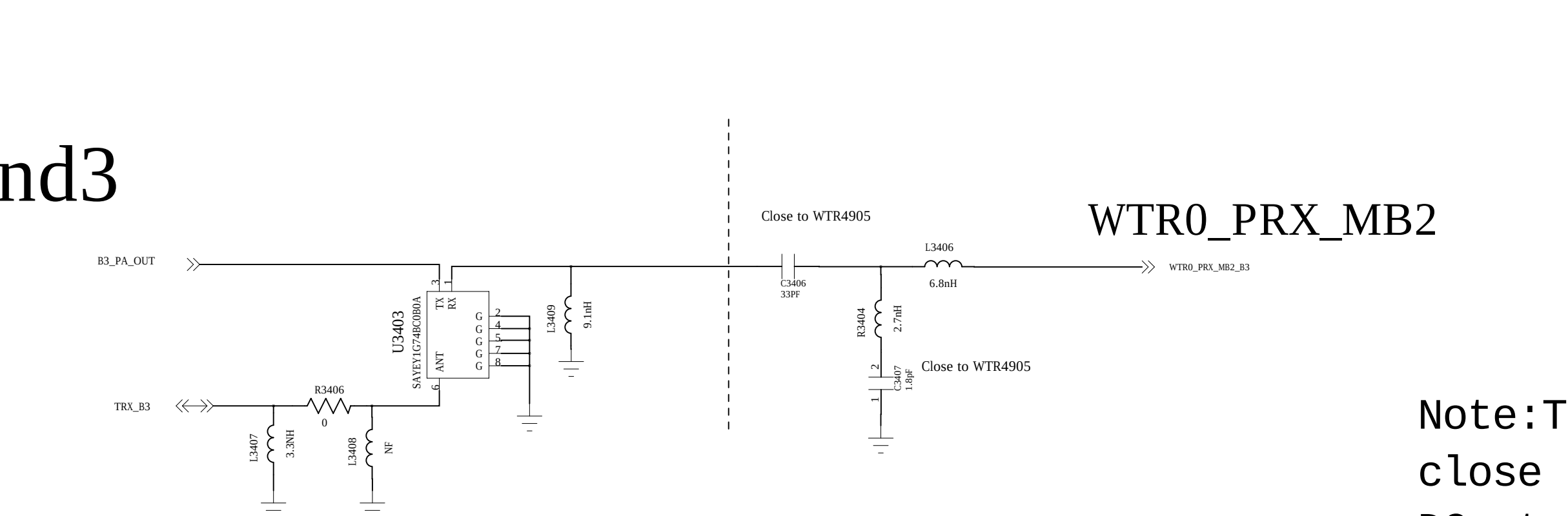
Band2



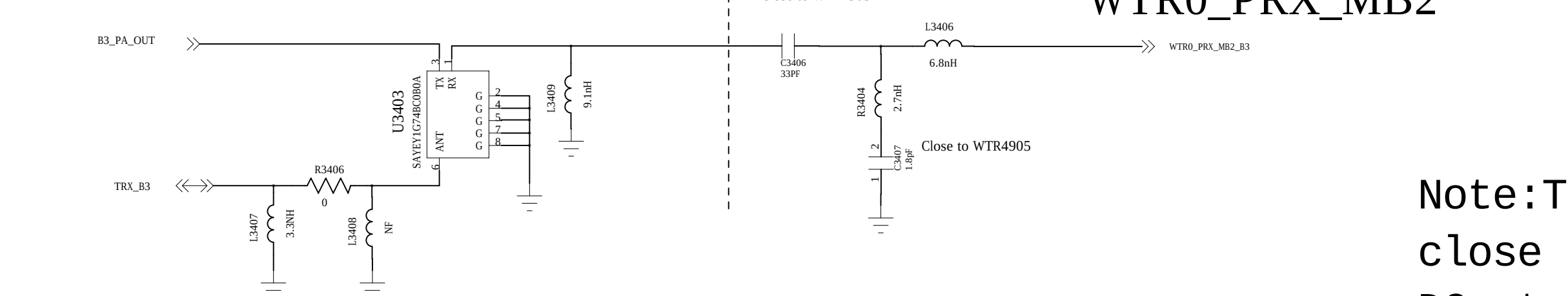
Band39



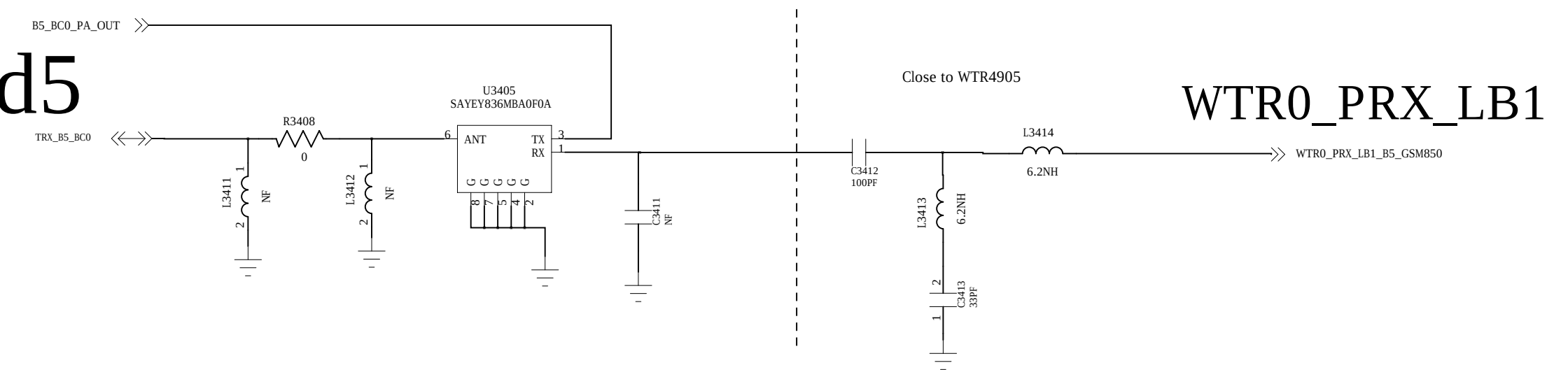
Band34



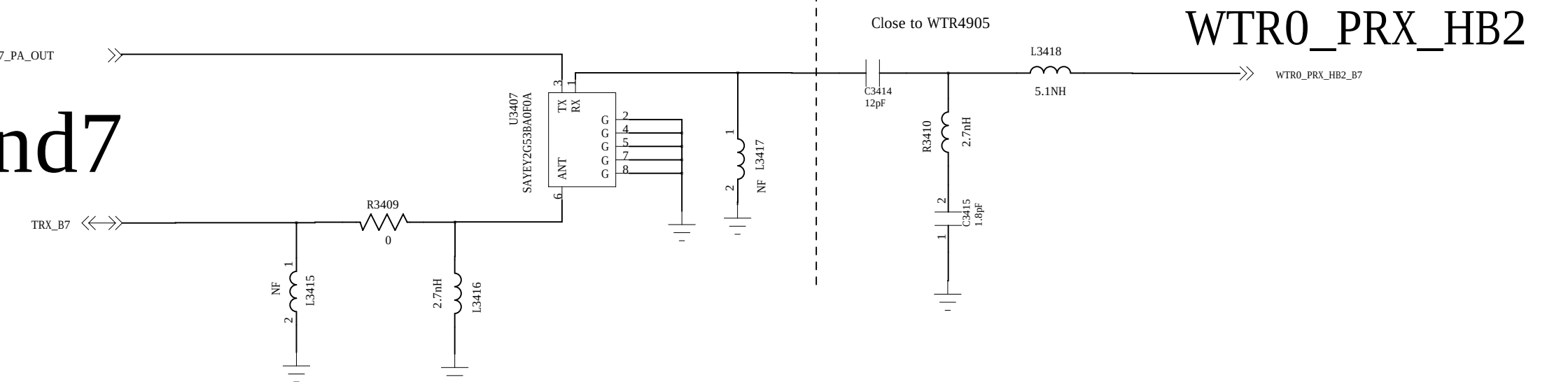
Band3



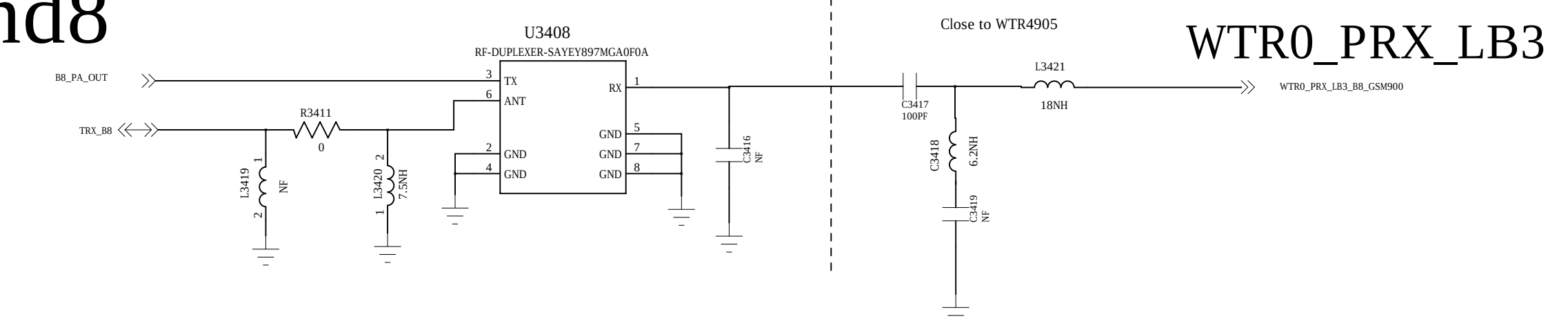
Band5



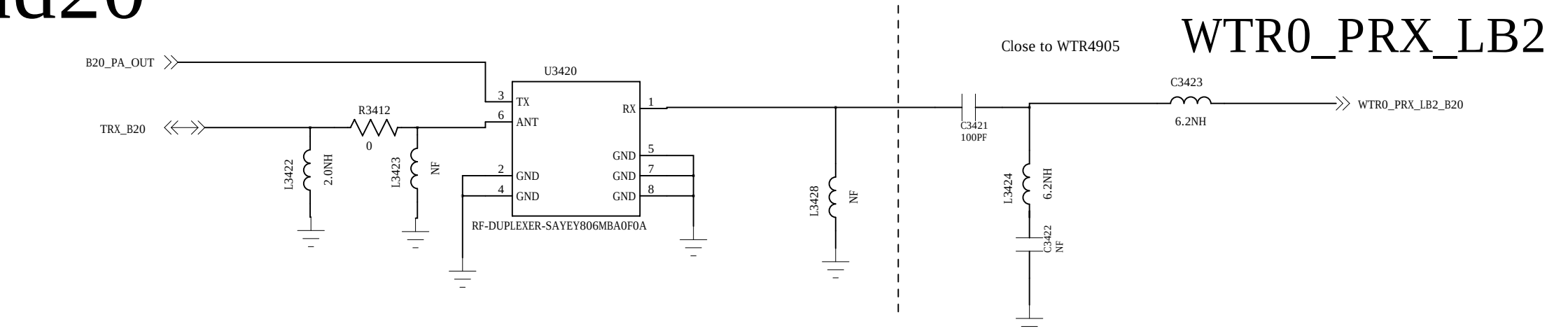
Band7



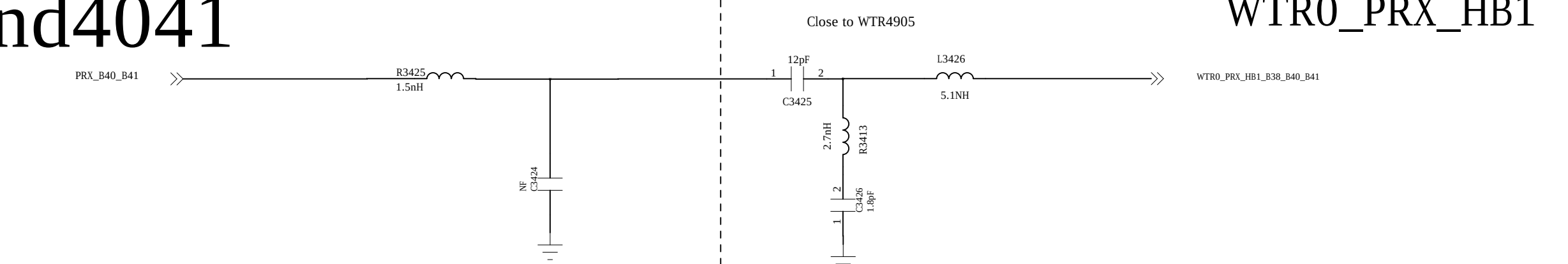
Band8



Band20

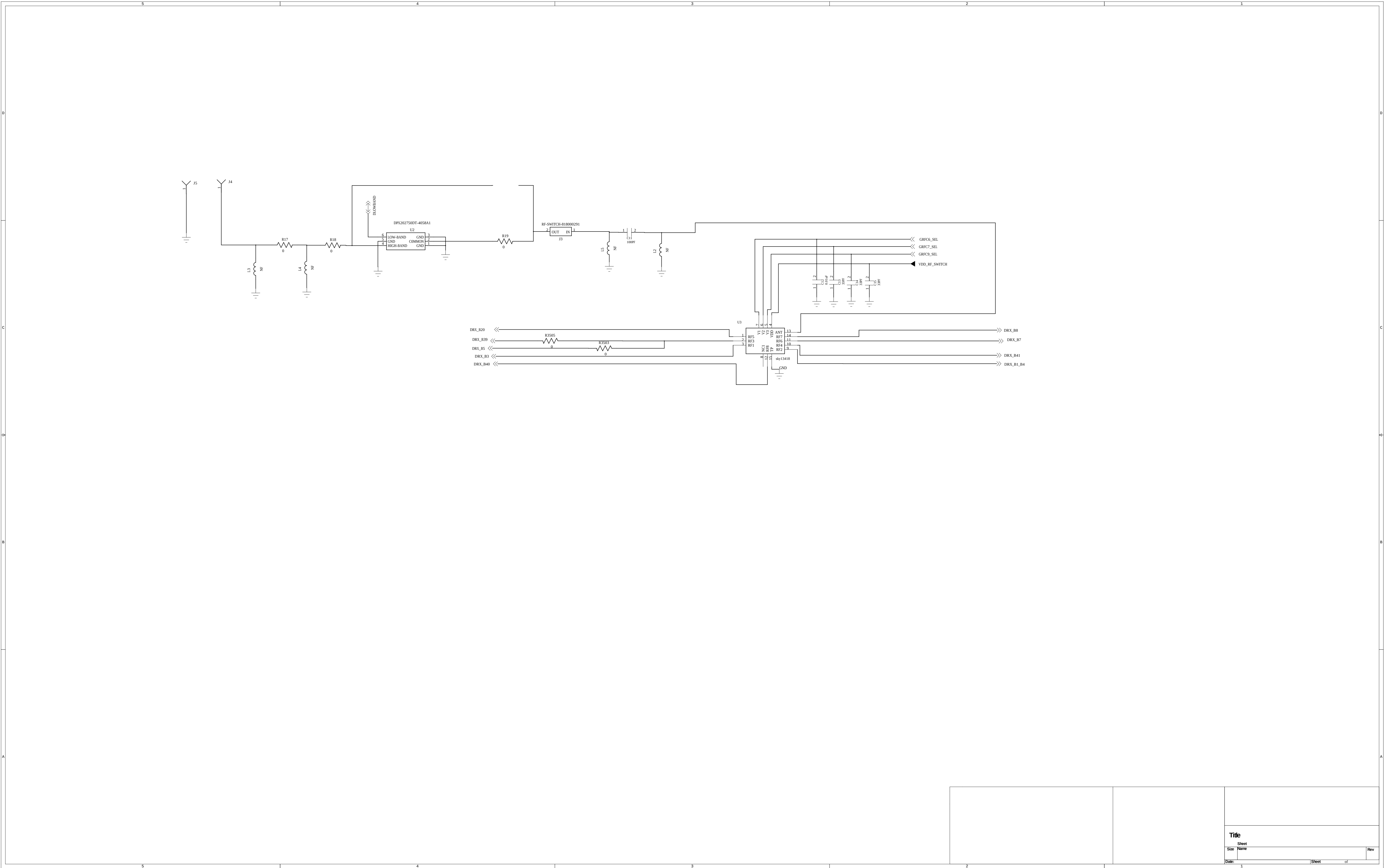


Band4041



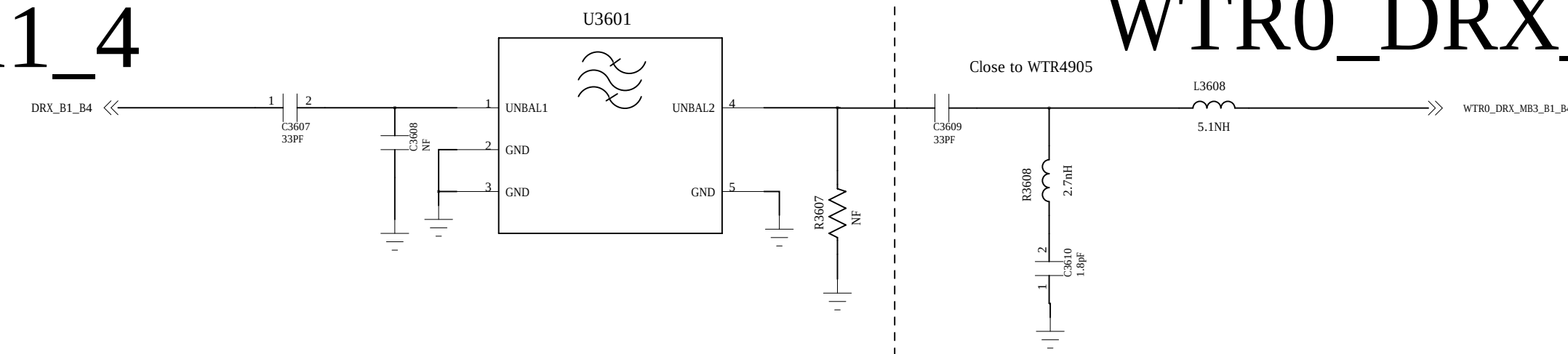
Note: The matching need close to WTR, RX ports have DC at the pin, so it need DC block,

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Sheet		
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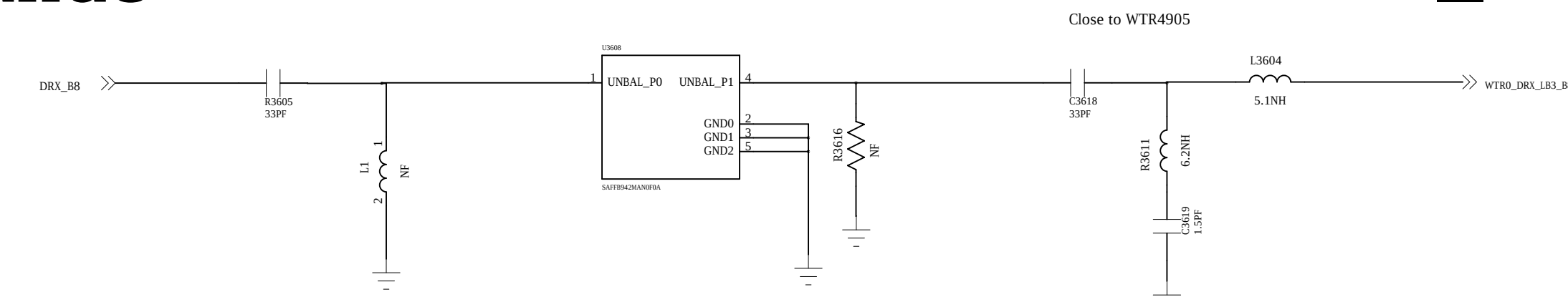
Band1_4

WTR0_DRX_MB3



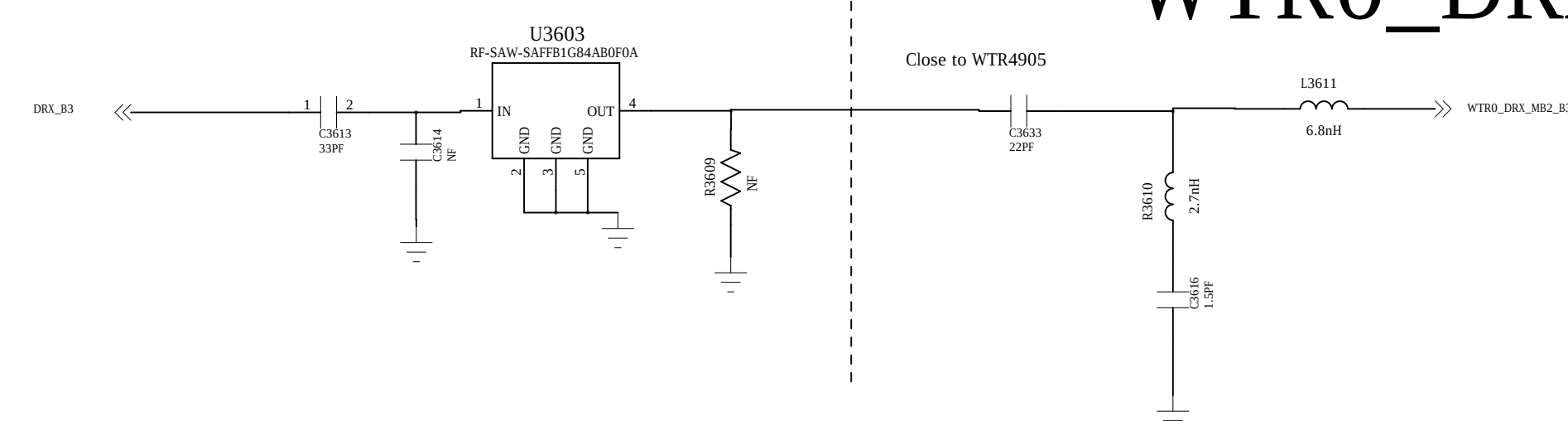
Band8

WTR0_DRX_LB3



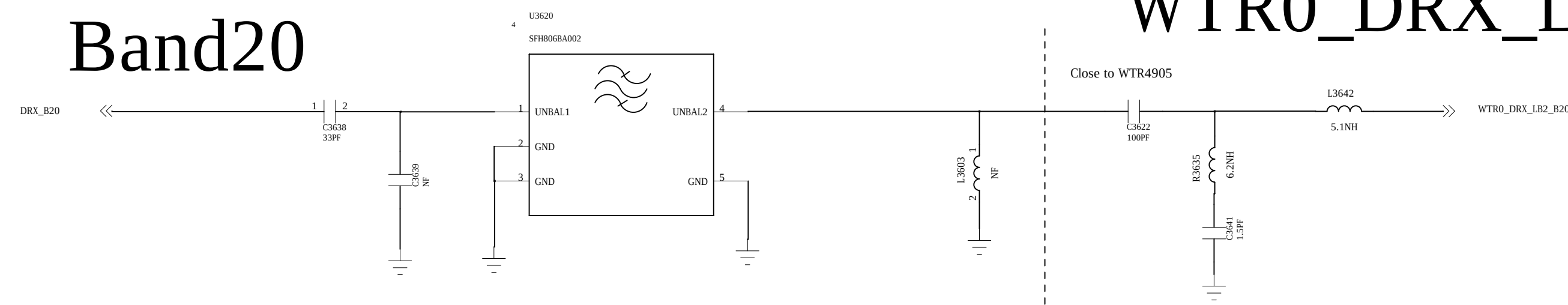
Band3

WTR0_DRX_MB2



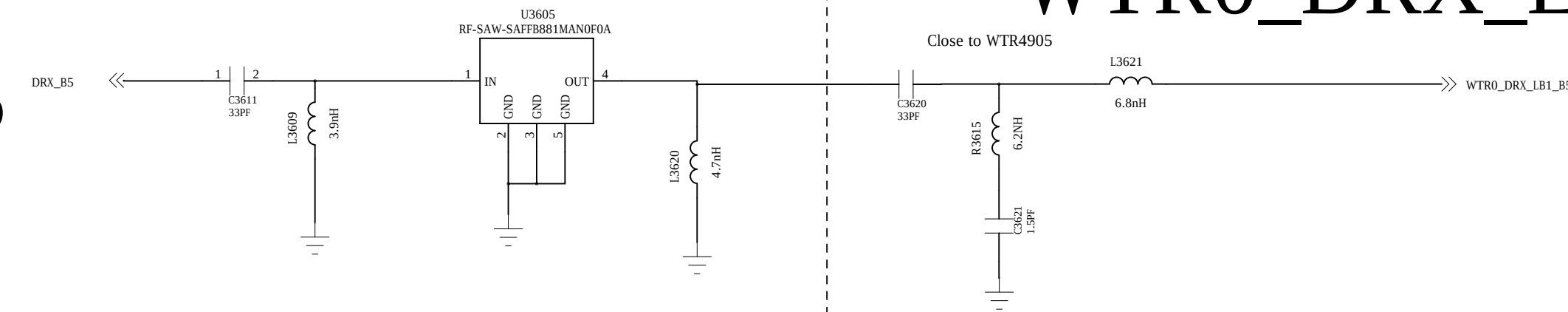
Band20

WTR0_DRX_LB2



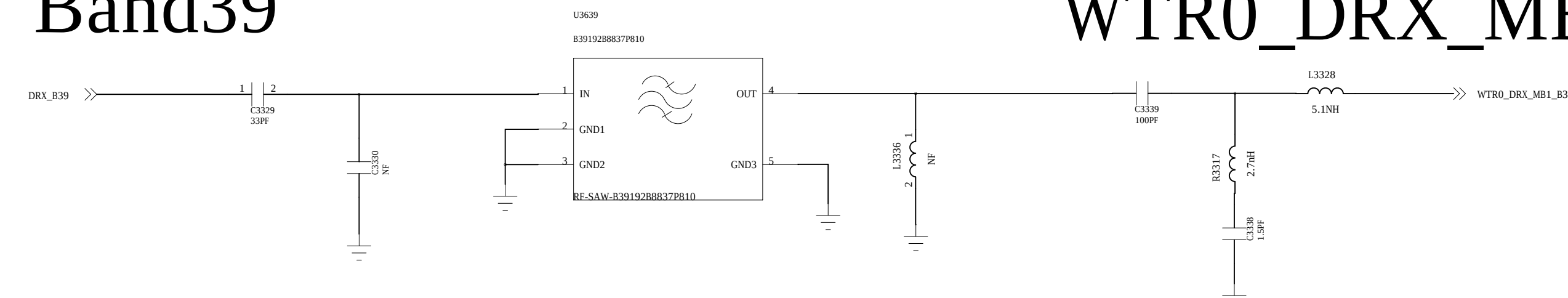
Band5

WTR0_DRX_LB1



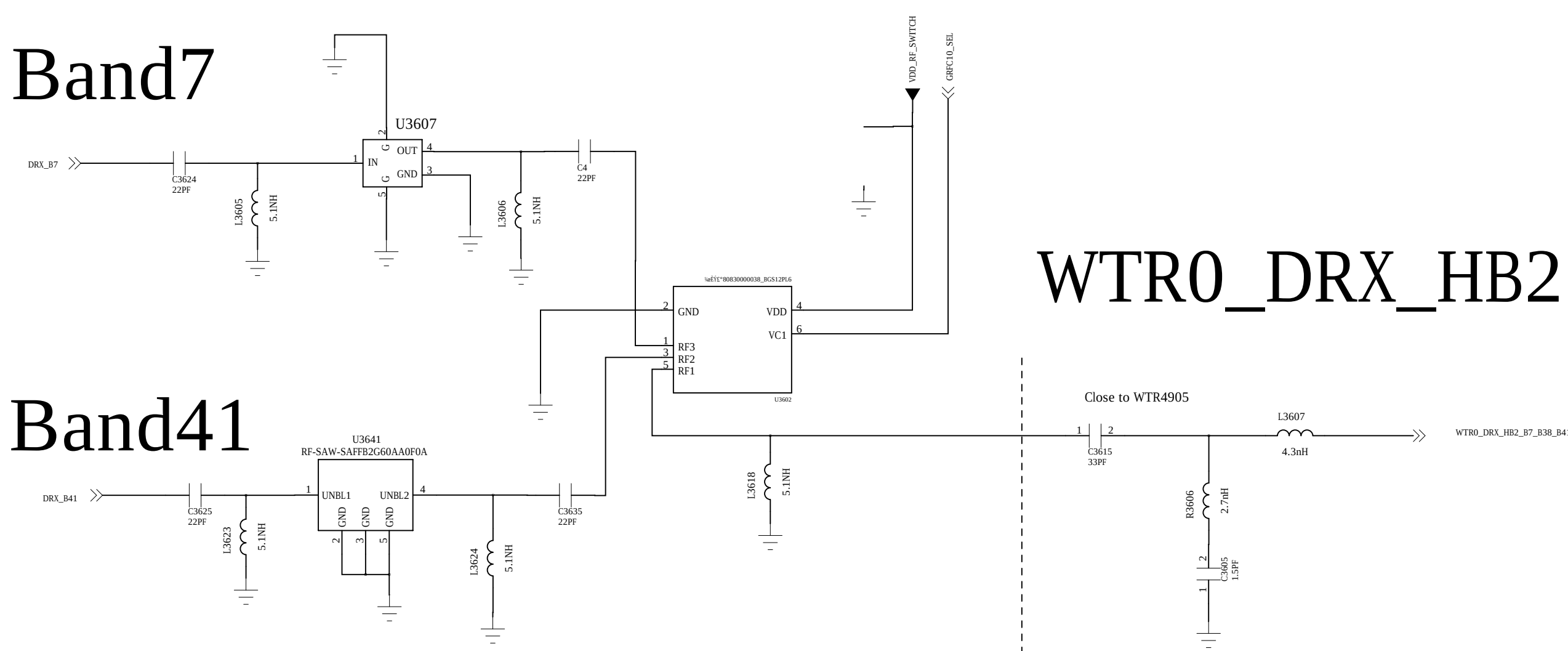
Band39

WTR0_DRX_MB1



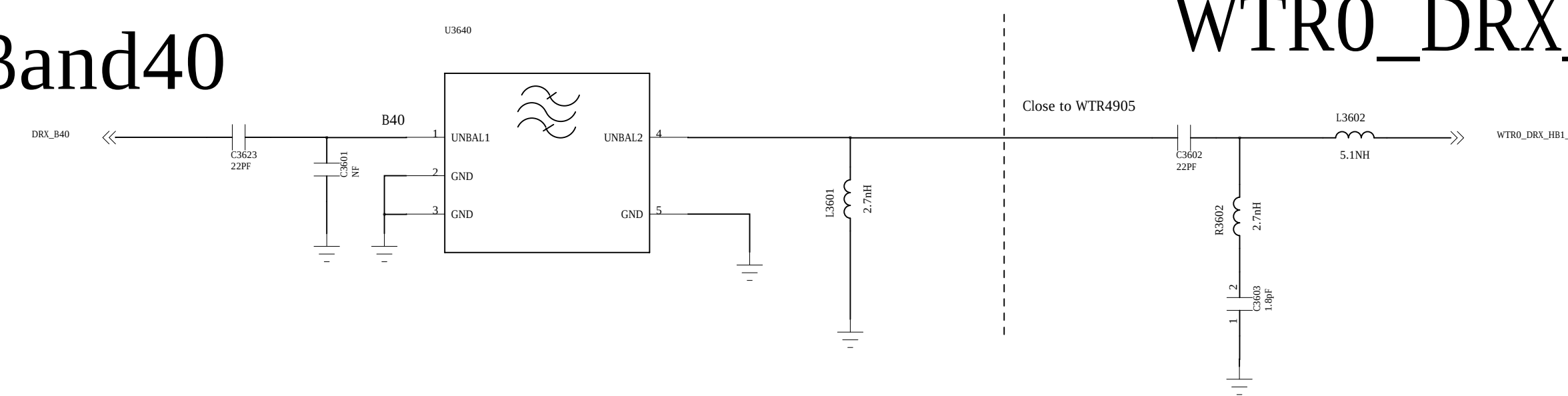
Band7

WTR0_DRX_HB2



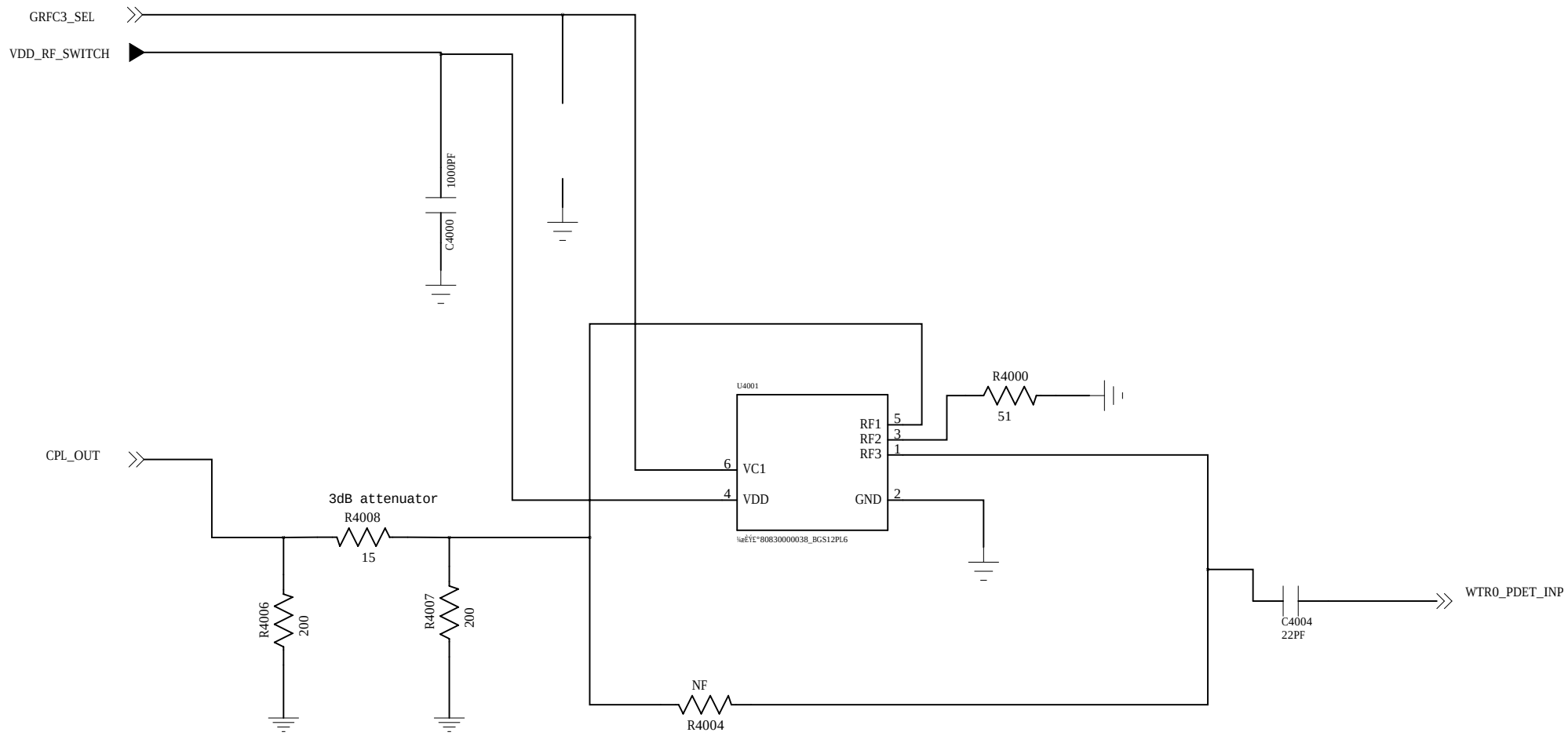
Band40

WTR0_DRX_HB1



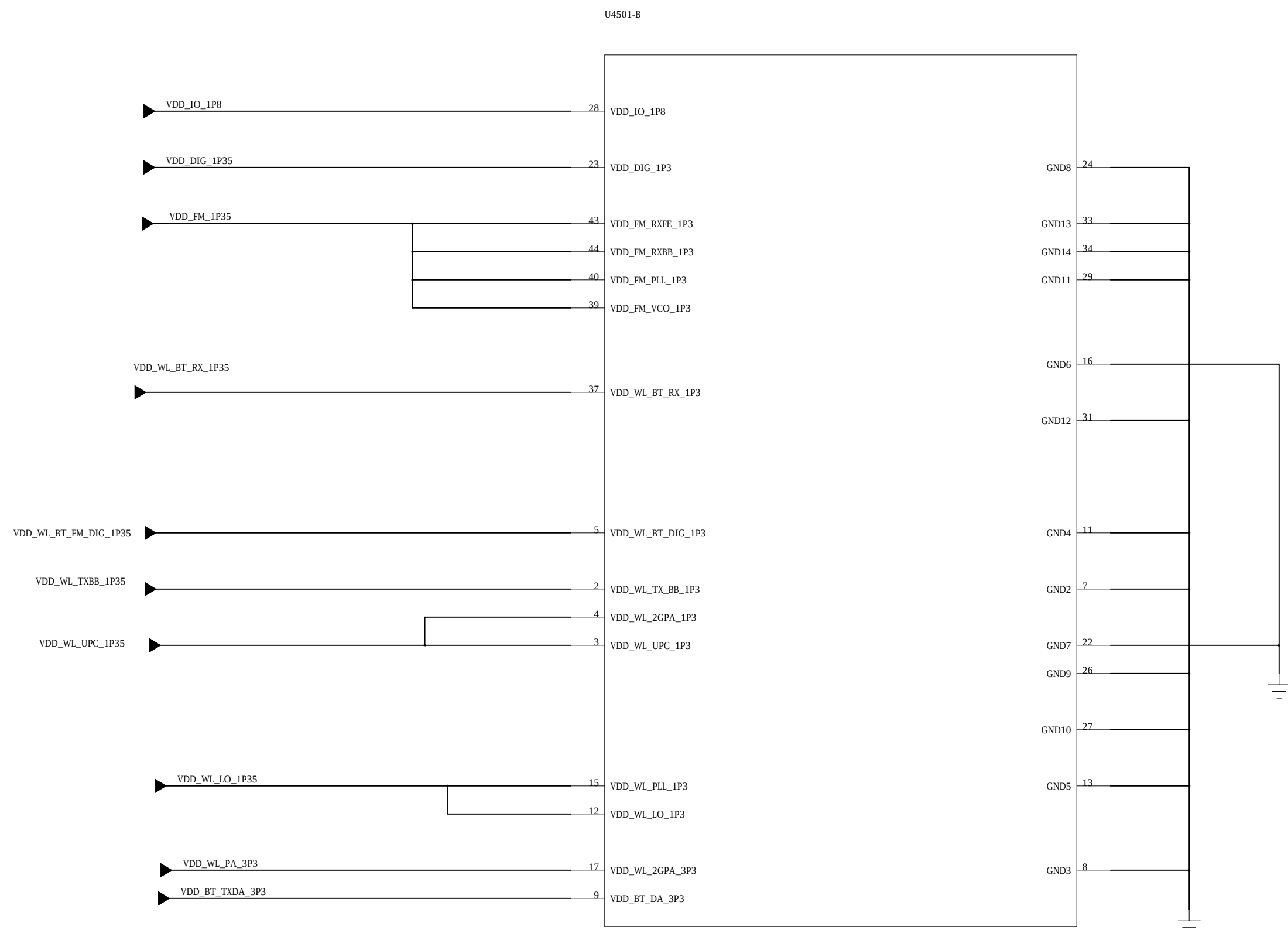
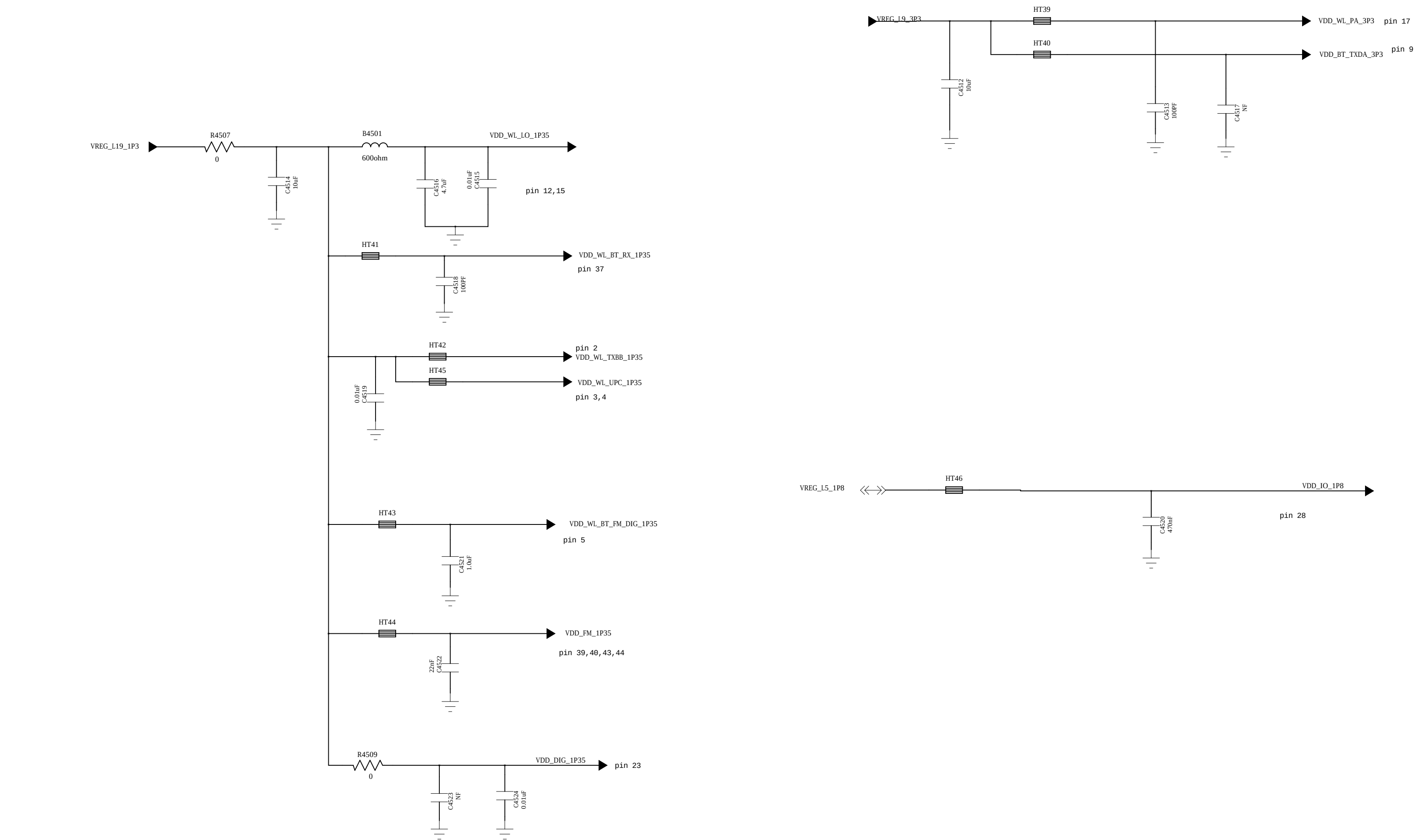
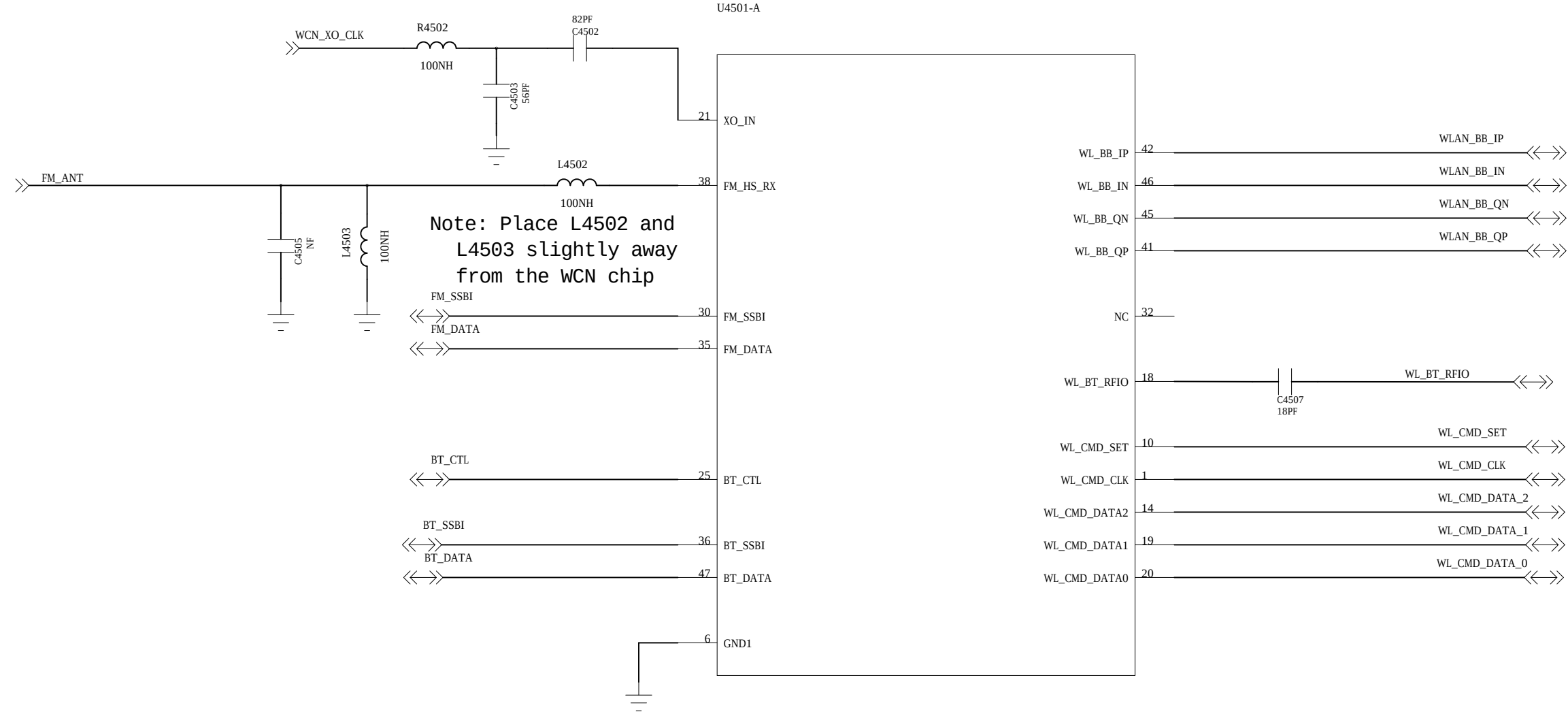
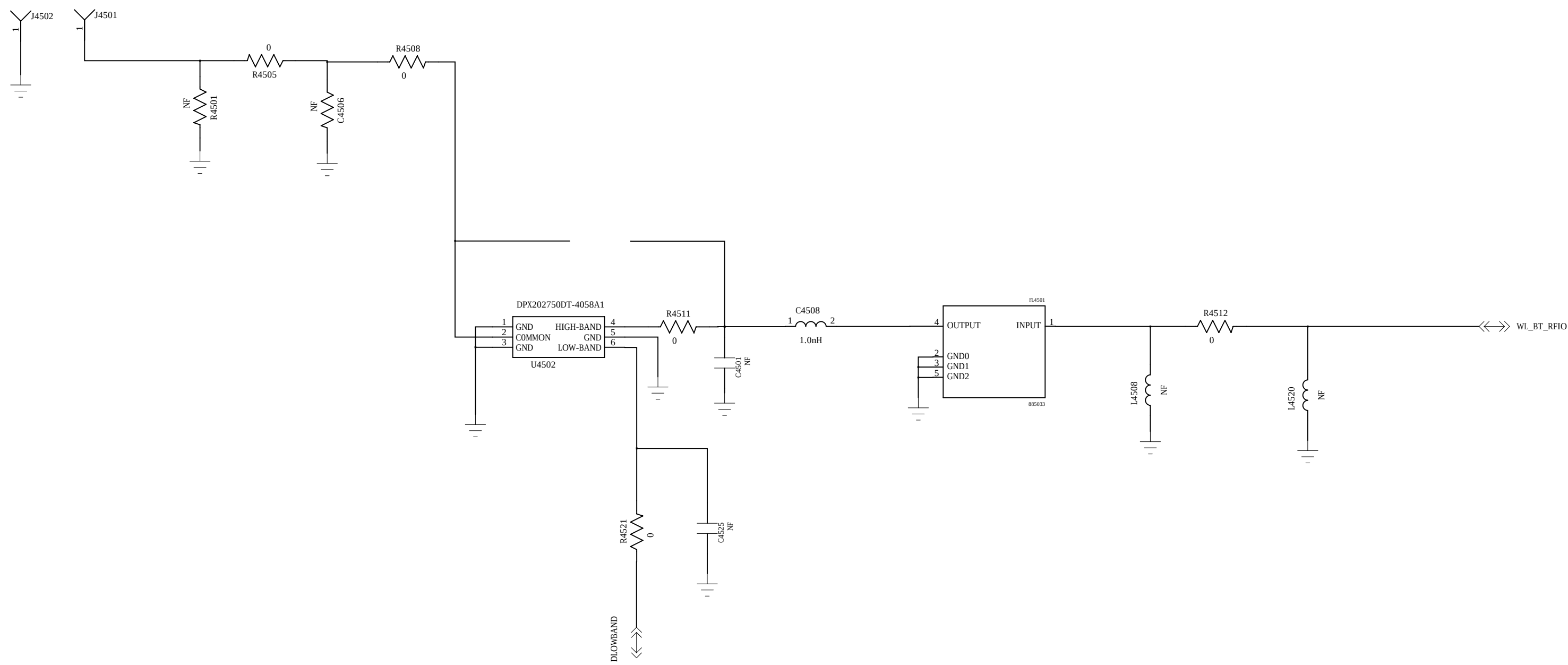
Note:The matching need close to WTR, RX ports have DC at the pin, so it need DC block,

SPDT



5	4	3	2	1						
Note: Use a signle low impedance power plane/fill VPA for all ET PA VCC1/VCC2. Refer to 80-NA681-91 rev.B or later revisions for more layout details.										
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Title										
Size	Name									
Date:	Sheet of									
5	4	3	2	1						

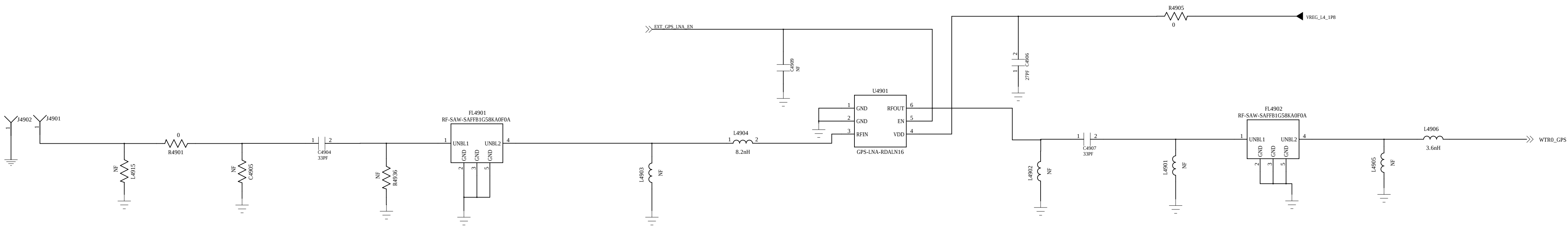
WCN



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